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Overview of Stark Industries Launch Systems



THRUST ISSUES

Juan Álvaro Cobos Franco

Mario Guida

Matteo Mafrici

Martina Lucia Magarelli

Camilla Martino

Margherita Palitta

Jules Jean Laurence Simon

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Abstract

This report presents the preliminary design and performance assessment of a Solid Rocket Motor (SRM) developed within the framework of a space propulsion project. The study focuses on the definition of a Ballistic Test and Evaluation System (BATES) grain configuration and a conical nozzle capable of delivering a nominal thrust of 100 kN under controlled operating conditions. The propellant selection, based on Ammonium Perchlorate (AP)/Hydroxyl-Terminated Polybutadiene (HTPB), is supported by a thermochemical analysis aimed at optimizing the oxidizer-to-fuel ratio while maintaining acceptable thermal loads.

An iterative optimization procedure is implemented to determine the grain geometry that ensures a near-neutral thrust profile, satisfying constraints on chamber pressure, burning time, and total impulse. The nozzle is designed consistently with the grain solution, prioritizing experimental feasibility and thermal analysis over maximum performance efficiency.

Particular attention is devoted to thermal protection systems, including the selection and sizing of thermal barrier coatings (Thermal Barrier Coatings (TBCs)) for both the combustion chamber and nozzle. The analysis highlights the trade-offs between material properties, thermal resistance, and structural limitations. A transient internal ballistics model, coupled with a Chemical Equilibrium with Applications (CEA)-based thermochemical solver, is used to evaluate real motor performance, accounting for efficiency losses and time-dependent effects.

Additional investigations include erosive burning assessment and a Monte Carlo analysis to quantify the impact of uncertainties on key performance parameters such as chamber pressure and burn time. Results demonstrate that the proposed design meets the mission requirements with limited deviations, confirming its robustness within the adopted modeling assumptions and providing a solid basis for further refinement.

Nomenclature

α	Coefficient of Thermal Expansion	$[\text{K}^{-1}]$
α_d	Divergent Half-Angle	$[\text{deg}]$
β	Convergent Half-Angle	$[\text{deg}]$
$\Delta p_{c,pp}$	Peak-to-Peak Pressure Variation	$[\text{bar}]$
ΔT	Temperature Gradient	$[\text{K}]$
δ_{met}	Metal Thickness	$[\text{m}]$
δ_{TBC}	Thermal Barrier Coating (TBC) Thickness	$[\text{m}]$
$\dot{m}$	Mass Flow Rate	$[\text{kg/s}]$
ϵ	Nozzle Expansion Ratio	$[-]$
ϵ_{flat}	Pressure Flatness Index error	$[-]$
ϵ_{m_p}	Relative Error on Propellant Mass	$[-]$
ϵ_{p_c}	Relative Error on Mean Chamber Pressure	$[-]$
ϵ_{t_b}	Relative Error on Burning Time	$[-]$
η_{c^*}	Characteristic Velocity Efficiency	$[-]$
η_{C_f}	Thrust Coefficient Efficiency	$[-]$
γ	Heat Capacity Ratio	$[-]$
λ_{conical}	Conical Nozzle Loss Factor	$[-]$
$\mathcal{N}(0, 1)$	Standard Normal Random Variable	$[-]$
μ	Statistical Mean	$[-]$
ρ	Density	$[\text{kg/m}^3]$
ρ_f	Throat Fillet Radius	$[\text{m}]$
σ	Statistical Standard Deviation	$[-]$
σ_ρ	Standard Deviation of Propellant Density	$[\text{kg/m}^3]$
σ_a	Standard Deviation of Pre-Exponential Factor	$[\text{mm}/(\text{s} \cdot \text{bar}^n)]$

σ_n	Standard Deviation of Pressure Exponent	[-]
σ_{amm}	Allowable Stress	[Pa]
σ_{c^*}	Standard Deviation of Characteristic Velocity	[m/s]
τ_{bo}	Burnout Time Constant	[s]
τ_{ign}	Ignition Time Constant	[s]
a	Pre-exponential Factor	[mm/(s · bar ⁿ)]
A_b	Area of Burning Surface	[m ²]
$A_{lat,conv}$	Convergent Section Lateral Area	[m ²]
$A_{lat,div}$	Divergent Section Lateral Area	[m ²]
$A_{lat,fillet}$	Throat Fillet Lateral Area	[m ²]
A_p	Area of the Port	[m ²]
A_t	Area of the Throat	[m ²]
Bi	Biot Number	[-]
C	Thermal Capacity	[J/K]
c^*	Characteristic Velocity	[m/s]
c_p	Specific Heat	[J/(kg · K)]
D	Outer Diameter of the Grain	[m]
d	Thickness of the Thermal Protection	[m]
d_0	Initial Port Diameter	[m]
d_e	Exit Diameter	[m]
d_t	Throat Diameter	[m]
E	Elastic Modulus	[Pa]
E	Weld Joint Efficiency	[-]
f	Friction Coefficient	[-]
FS	Safety Factor	[-]
G	Hot Gas Mass Flux Over Burning Surface	[kg/(m ² · s)]
h	Convective Heat Coefficient	[W/(m ² · K)]
h_{air}	Air Convective Coefficient	[W/(m ² · K)]
h_g	Gas Convective Coefficient	[W/(m ² · K)]

I_{sp}	Specific Impulse	[s]
I_{tot}	Total Impulse	[Ns]
J	Optimization Cost Function	[-]
K	Minor Loss Coefficient	[-]
k	Thermal Conductivity	[J/(m · K · s)]
K_n	Area Ratio	[-]
K_n	Klemmung (Burning-to-Throat Area Ratio)	[-]
K_{IC}	Fracture Toughness	[MPa · m ^{1/2}]
k_{met}	Metal Conductivity	[W/(m · K)]
k_{TBC}	Thermal Conductivity of the TBC	[W/(m · K)]
L	Characteristic Length of the Combustion Chamber	[m]
L_0	Segment Length	[m]
L_{conv}	Convergent Section Length	[m]
L_{div}	Divergent Section Length	[m]
L_{noz}	Total Nozzle Length	[m]
M	Mach Number	[-]
N	Number of BATES Segments	[-]
n	Pressure Exponent	[-]
n_p	Polytropic Exponent	[-]
N_{mc}	Number of Monte Carlo Realizations	[-]
Nu	Nusselt Number	[-]
O/F	Oxidizer-to-Fuel Ratio	[-]
p_c	Combustion Chamber Pressure	[bar]
$P_{c,MEOP}$	Maximum Expected Operating Pressure	[bar]
Pr	Prandtl Number	[-]
q''	Convective Heat Flux	[W/m ²]
r_b	Burning Rate	[mm/s]
r_0	Non-erosive Burning Rate Contribution	[mm/s]
$r_{b,target}$	Target Burning Rate	[mm/s]

r_e	Erosive Burning Rate Contribution	[mm/s]
Re	Reynolds Number	[-]
T	Thrust	[N]
t	Time	[s]
t_b	Burning Time	[s]
T_c	Combustion Chamber Temperature	[K]
T_i	Initial Temperature	[K]
T_{air}	Air Temperature	[K]
T_{aw}	Adiabatic Wall Temperature	[K]
$t_{b,target}$	Target Burning Time	[s]
$T_{external}$	External Temperature of the Case	[K]
T_{int}	Interface Temperature	[K]
T_{w1}	Temperature of the Superior Wall of the Topcoat	[K]
T_w	Wall Temperature	[K]
w	Web Thickness	[m]
w_{ideal}	Ideal Web Thickness	[m]
x	Spatial Coordinate	[m]

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List of Acronyms

1D One-Dimensional.

AP Ammonium Perchlorate.

APS Atmospheric Plasma Spray.

BATES Ballistic Test and Evaluation System.

BM Bates Motors.

C/C Carbon-Carbon Composite.

CEA Chemical Equilibrium with Applications.

CFD Computational Fluid Dynamics.

DCL Double Ceramic Layer.

EB-PVD Electron Beam Physical Vapour Deposition.

FAC Finite Area Combustor.

GZO $Gd_2Zr_2O_7$, Gadolinium Zirconate.

HTPB Hydroxyl-Terminated Polybutadiene.

IAC Infinite Area Combustor.

LTM Low Thermal Conductivity.

LZO $La_2Zr_2O_7$, Lanthanum Zirconate.

MEOP Maximum Expected Operating Pressure.

NASA National Aeronautics and Space Administration.

PS-PVD Plasma Spray Physical Vapour Deposition.

RGO Rapid Growth of Oxide.

RSZ Rare-earth Stabilized Zirconia.

SIL Software-in-the-Loop.

SRM Solid Rocket Motor.

TBC Thermal Barrier Coating.

TBCs Thermal Barrier Coatings.

TGO Thermally Grown Oxide.

YSZ Yttria-stabilized Zirconia.

Chapter 1

State of the Art

1.1 Introduction

The constantly evolving world of Space Propulsion requires its engineers to delve into physics and chemistry to design extraordinary yet cost efficient solutions, for mankind to adventure in a such extreme adverse environment. This paragraph aims to analyze the studies and conclusions shown in the given literature and beyond, regarding TBCs, with particular attention to latest advancements in RSZ/YSZ layers, as well as Bates Motors (BM), to eventually be considered for the sizing part of the company's new Stark-100k.

1.2 Thermal Barrier Coatings

TBCs are multi-layered ceramic systems which have been in operational use on gas turbine hot-section components since the 70s. During the last 20 years of development [1, 2], new material families emerged, along with advanced deposition technologies. Such improvements were brought to the reusable rocket engines field, which requires temperatures far beyond those used in initial studies. Above 1200°C, YSZ undergoes unfavorable phase transformations, causing volumetric changes and crack formation within coating, thus a double layered system of RSZ/YSZ has been developed to satisfy the market request.

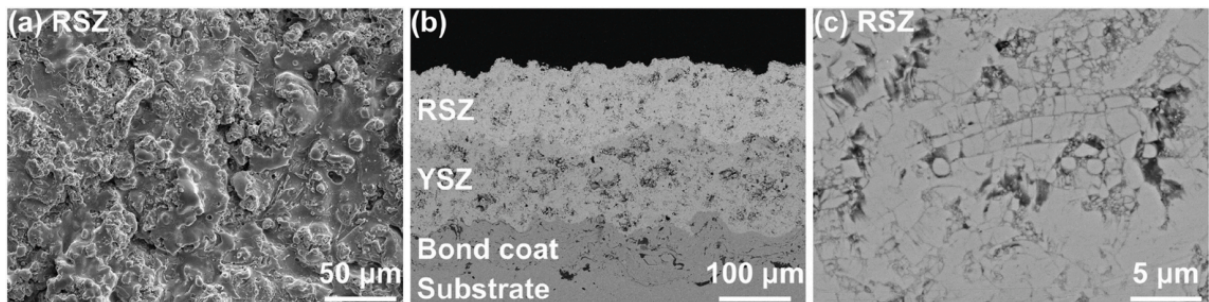


Figure 1.1: Surface (a), cross-section (b,c) of microstructures APS RSZ/YSZ coatings [3]

1.2.1 Performance and criticalities

The performance of a thermal barrier coating is governed by four mutually coupled properties that must be simultaneously optimised: Low Thermal Conductivity (LTM) to maximise the temperature drop across the coating; high fracture toughness to resist crack

propagation under thermal shock; a coefficient of thermal expansion close to that of the metallic substrate to limit thermomechanical mismatch stresses; high infrared reflectance, crucially at temperatures above 1600°C, to prevent radiative heat penetration through the ceramic layer to the substrate. No single material excels in all four simultaneously, which motivates the use of Double Ceramic Layer (DCL) architectures, where a topcoat with superior high-temperature properties is deposited over a YSZ interlayer that acts as chemical buffer between the topcoat and the bond coat. However, in order to further increase the operating temperature, some new ceramic oxides with distinctly LTM, high phase stability and high sintering resistance (other properties related to capacity of remaining stable under varying conditions and of resisting contraction due to cracks to high temperature exposure) have great potentials as next generation of high performance TBCs topcoat materials: in particular, rare-earth zirconates of the general form $Ln_2Zr_2O_7$ (like $La_2Zr_2O_7$, Lanthanum Zirconate (LZO) and $Gd_2Zr_2O_7$, Gadolinium Zirconate (GZO)) with a pyrochlore-type or a defect fluorite-type structure show promising thermophysical properties that have the phase transition temperature higher than that of YSZ. However, rare-earth zirconates have lower thermal expansion coefficients than the bond coat, which will result in higher thermal stress in the TBCs systems, so their thermal cycling properties are worse than that of YSZ [4] and their price is higher.

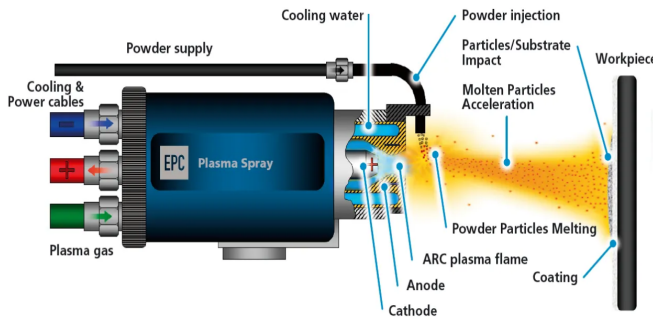


Figure 1.2: Schematic of APS system [5]

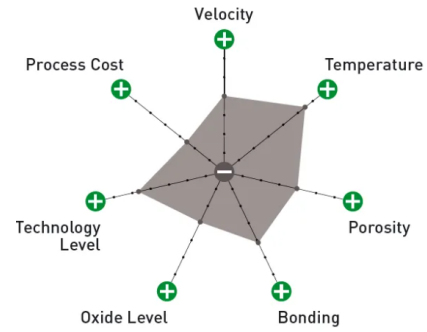


Figure 1.3: APS performance chart [5]

Generally, Electron Beam Physical Vapour Deposition (EB-PVD) and APS are the two main methods of depositing TBCs. Among the main advantages of APS are the possibility to use different mixtures for coating (also among the ones just mentioned), without having to modify the technology and procedure. Needless to say, industrialized procedures are the baseline strategy for cheaper and faster production chains. While other deposition technologies, such as EB-PVD and Plasma Spray Physical Vapour Deposition (PS-PVD) require high vacuum infrastructure and/or preheats procedures of the substrate, working at atmospheric pressure results also in a ground-testing facility advantage. In addition, APS can quickly produce coatings with large thickness and good thermal insulation, but poor thermal shock resistance due to its lamellar structure; in contrast, EB-PVD exhibit high-strain tolerance thanks to its columnar structure, but inadequate thermal insulation [6, 7].

1.3 BATES motor

Efficiency and performance shall always meet with the capability to build and test on ground smaller and bigger scaled down components, thus our necessity to conduct tests on BATES motors. The first BATES motor was fired on 8 September 1961 after the Air

Force Rocket Propulsion Laboratory wished to establish a high-precision ballistic testing capability that could rank vendors propellant performance and provide an estimate of full-scale motor performance using the minimum motor size possible to limit the quantity of scarce advanced propellants needed. There was the understanding of the need for a trace as neutral as possible and the minimization of heat-loss and mass addition in the motor design. The first design was of the 31.75 kg BATES motor. After that, there was the design of the 6.80 kg BATES motor and ultimately there was the Super BATES motor 362.87 kg and more propellant was added to approximate the gas and particle residence time of a large motor. In modern usage, BATES refers to a type of solid-fuel rocket motor grain geometry designed to perform neutral burning. These grains consist of one or more cylindrical propellant segments with a central axial perforation so that during the combustion the core diameter increases while the segment length decreases. With these effects balancing each other, the burning area remains nearly constant. This leads to a neutral thrust behavior, where chamber pressure and thrust remain approximately steady over time [8].

1.3.1 BATES grain configuration

A BATES grain configuration is typically used when a nearly neutral K_n profile is desired. K_n slope rises to a maximum value then. The shape of the curve is determined by the L_0/D and d_0/D ratios. Ideally, $K_{n,max}$ should occur at a surface regression of one-half the web thickness to produce a symmetric profile ($K_{n,in} = K_{n,fin}$)[9].

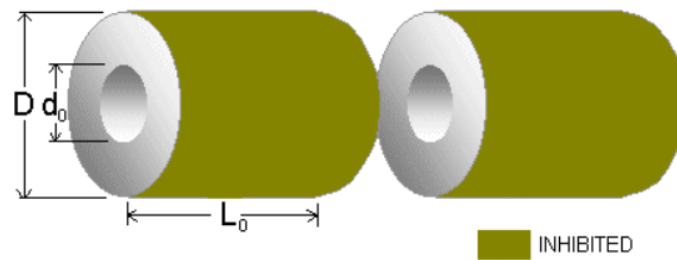


Figure 1.4: BATES [9]

A careful selection of the core diameter and segment length is required in order to properly shape the K_n evolution during combustion. If these geometric parameters are not appropriately balanced, the resulting K_n profile may deviate from the desired behavior and exhibit either a purely progressive trend, characterized by a monotonic increase of K_n , or a purely regressive trend, in which K_n decreases continuously through the burn [9].

Chapter 2

Propellant and Ballistic Properties Characterization

2.1 Propellant Selection

The assignment specifies the use of a standard non-aluminized AP/HTPB composite propellant with a maximum accepted chamber pressure of $p_c = 70$ bar. Within this family, the key design degree of freedom is the oxidizer-to-fuel mass ratio O/F , which governs simultaneously the thermochemical performance (characteristic velocity c^* , specific impulse I_{sp} , and adiabatic flame temperature T_c) and the severity of the thermal environment experienced by the motor casing, the nozzle throat, and the thermal protection system.

2.2 Expansion model and computation

The frozen flow assumption is applied from the throat onwards, meaning that the chemical composition of the combustion products is fixed at the throat section ($M = 1$) and does not evolve further through the divergent portion of the nozzle. From a performance standpoint, such choice yields lower predicted values of c^* and I_{sp} compared to the shifting equilibrium model, since it neglects the additional enthalpy release from recombination reactions occurring during expansion. This creates a conservative baseline: any real performance will be equal to or greater than the frozen prediction, providing a built-in safety margin in the sizing of the grain and nozzle. The difference between the two models for AP/HTPB propellants at 65 bar is typically in the range of 2–5 s [10] in specific impulse, which is non-negligible but acceptable at this stage of design, where uncertainties on propellant composition and combustion efficiency already dominate the error budget [11]. The results are presented in Table 2.1. Increasing the O/F ratio improves performance metrics like the characteristic velocity c^* and the I_s but also the T_c . Beyond $O/F \approx 6$ the performance improvement becomes progressively marginal, while the thermal loads in the chamber and nozzle increase severely. Therefore, an $O/F = 6.0$ (85.7% AP mass fraction), was selected as the optimal engineering compromise. Although not the absolute thermochemical peak, it satisfies thrust and impulse requirements while maintaining manageable thermal conditions for the motor. Crucially, the primary goal of this test is to study nozzle throat heat transfer, which requires stable geometries and thermal boundaries. Pushing for a higher O/F would accelerate thermal stress, compromising test reliability. Furthermore, operating at a slightly lower T_c minimizes the required thickness of the RSZ

Table 2.1: CEA Parametric Sweep Results for AP/HTPB Propellant

O/F	%AP	$c^*(\text{m/s})$	$I_{sp}(s)$	$T_c(K)$
4.0	80.0	1430.1	227.4	2342.9
4.5	81.8	1463.1	233.6	2529.4
5.0	83.3	1487.4	238.1	2680.1
5.5	84.6	1505.2	242.9	2800.0
6.0	85.7	1518.1	244.0	2892.5
6.5	86.7	1526.8	245.4	2959.9
7.0	87.5	1531.5	246.3	3004.7
7.5	88.2	1532.4	246.4	3029.4
8.0	88.9	1529.6	246.0	3037.4

thermal protection layer and ensures overall mechanical robustness, as will be discussed further in Chapter 3.

2.3 Ballistic characterization

The ballistic properties of the propellant have been characterized considering fifteen experimental data relative to different BATES motors. The equation used for the evaluation of the ballistic properties is the Vieille’s law: $r_b = ap_c^n$. The analysis has been conducted performing a linear regression in order to obtain the coefficients a and n that are (1.6177 ± 0.0646) and (0.3816 ± 0.0106) , respectively; with a data fitting of 0.99, suggesting a deviation of 1%. So, the results can be considered close to the experimental ones. Since the pressure in the combustion chamber has been set to $p_c = 65$ bar, a burning rate has been evaluated for this value. The result is $r_b = 7.9548$ mm/s.

2.4 Grain And Nozzle Geometry Design

2.4.1 Design objectives and Methodology

The goal of this phase is to define a motor geometry capable of producing thrust $T = 100$ kN at sea level, with a total impulse of $I_{tot} = 2.5 \times 10^6$ N · s and an average chamber pressure of $p_c = 65$ bar. It is important to mention that the design of the grain is driven by the requirement for a neutral thrust profile to ensure steady state conditions for the nozzle throat heat transfer characterization. Specifically, in a BATES grain, the total burning Area (A_b) is the sum of the internal cylindrical surfaces (which increases with regression) and the annular end faces (which decreases with regression). A key engineering choice is taken for the length of the grain, to ensure these two effects balance each other effectively. This is achieved by enforcing the geometric constraint shown in Equation 2.1. Since the web thickness is defined by Equations (2.2) and (2.3), the constraint guarantees that the maximum burning area occurs exactly at the midpoint of the burn, leading to a symmetric, nearly neutral pressure trace [9]. Thus, Equation (2.1) is equivalent to Equation (2.3):

$$L_0 = \frac{1}{2}(3D + d_o) \quad (2.1) \qquad w = \frac{D - d_o}{2} \quad (2.2) \qquad L_0 = 2d_o + 3w \quad (2.3)$$

2.4.2 Iterative optimization loop

Because the regression rate is coupled to chamber pressure through Vieille’s law, the grain design cannot be finalized by a purely algebraic sizing. In fact, the burning rate depends on pressure according to the Vieille’s law while the chamber pressure itself depends on the instantaneous ratio between the burning area and the throat area. As a consequence, the grain geometry, the pressure evolution and the burning time are mutually coupled, and an iterative procedure is required to identify a configuration simultaneously satisfying thrust, impulse, pressure level, and neutrality requirements. The optimization was carried out through a parametric sweep on the two main grain design variables: the number of BATES segments N and the web thickness w . The final design was chosen as the configuration that minimized a weighted cost (or penalty), which is described later in this subsection. The first loop was performed on the number of segments, considering values in the range $N = 1 \dots 6$ in order to explore the trade-off between ballistic neutrality and overall compactness. In general, using fewer segments tends to produce a flatter pressure profile, while increasing N allows the same propellant mass into a shorter motor. This, however, increases the complexity of the design and augments the number of interfaces between segments. At this point, for each chosen value of N , a second sweep was carried out on the web thickness. An initial estimate of the required web was obtained from the target burn time as:

$$w_{\text{ideal}} = r_{b,\text{target}} t_{b,\text{target}} \quad (2.4)$$

The actual optimization was then carried out by varying w within a $\pm 15\%$ interval around this value. This refinement is necessary because the average burning rate during firing is not exactly equal to the burning rate computed at the nominal chamber pressure: even a limited change of p_c from the reference value modifies r_b , which in turn changes the effective burning time. Refer to Appendix A to see how actually an iteration behaves. For each candidate geometry, four performance indicators were evaluated: the estimated burn time, the total propellant mass, the average chamber pressure, and a flatness index defined as the peak-to-peak pressure variation normalized by the mean chamber pressure. These quantities were then compared with the target values through the following weighted cost function

$$J = 15 \varepsilon_{\text{flat}} + 10 \varepsilon_{p_c} + 10 \varepsilon_{t_b} + 5 \varepsilon_{m_p}, \quad (2.5)$$

where $\varepsilon_{\text{flat}}$ is the non-uniformity of the normalized pressure and ε_{p_c} , ε_{t_b} , and ε_{m_p} are the relative errors on mean chamber pressure, burning time, and propellant mass, respectively. The adopted weights reflect the design priority of preserving an almost neutral pressure trace while remaining close to the assigned operating point and impulse requirement. An additional soft penalty was included to discard solutions with excessively large outer diameters, even if they were otherwise acceptable. Thus, the optimal grain configuration was then selected as the one minimizing the cost function. Its geometry was subsequently used for the final nozzle sizing and for the time-resolved performance analysis presented in the following sections.

2.4.3 Optimized Geometry

The optimization procedure described in section 2.4.2 led to a single grain configuration that minimizes the weighted cost function while maintaining a nearly neutral pressure

evolution. The selected solutions meet the target operating conditions with only small deviations in average chamber pressure and burn time, and keeps the peak-to-peak pressure variation low over the entire burn. It was therefore adopted as the reference geometry for the time resolved performance analysis that follows. The optimized BATES grain geometry is presented in Table 2.2. The selected geometry represents the best compromise between pressure-trace flatness, propellant mass consistency, and overall geometric feasibility.

Table 2.2: Optimized BATES grain geometry

Parameter	Value	Parameter	Value
Number of segments, N	1	Max K_n [-]	322.625
Outer diameter, D [m]	0.8714	Propellant Mass M_p [kg]	1093.22
Initial core diameter, d_0 [m]	0.4738	Est. burning time, t_b [s]	24.99
Segment length, L_0 [m]	1.5441	Avg chamber pressure, $\bar{p}_c$ [bar]	64.98
Web thickness, w [m]	0.1988	P-to-P variation, $\Delta p_{c,pp}$ [bar]	5.96
Initial burning area, $A_{b,0}$ [m ²]	3.1386	Pressure flatness index [%]	9.18
Initial K_n [-]	304.551		

From the optimized grain solution, the geometry of the nozzle was sized by imposing the throat area A_t required by the target mass flow rate and by applying the selected expansion ratio retrieved from National Aeronautics and Space Administration (NASA) CEA $\varepsilon = 8.85$. Although Rao nozzles are generally preferred in applications where maximum thrust efficiency is a primary objective, a conical nozzle was deliberately selected for the present system. This choice was driven by the specific experimental purpose of the motor firing. In fact, the assignment is primarily focused on the quantification of thermal loads in the throat region through a reusable straight cooling jacket, rather than on the maximization of nozzle performance. Under these conditions, a conical nozzle offers a substantial practical advantage: its linear shape leads to straight external walls, which are inherently more compatible with a straight cooling jacket and allow a simpler, more predictable coupling between the nozzle wall and the cooling-water passage. By contrast, a Rao contour would require either a shaped cooling channel or locally varying contact conditions, thus complicating both the manufacturing process and the interpretation of the thermal experiment. A second reason for retaining the conical solution lies in its simplicity of manufacture. The modest performance penalty associated with a conical nozzle is considered an acceptable trade-off in exchange for reduced machining complexity, lower cost, and greater robustness of the experimental setup. The conical geometry therefore represents the most coherent solution for the present pre-design and test-oriented activity. The convergent and divergent sections were reconstructed directly from the imposed half-angles, selected according to standard solid-rocket preliminary-design practice. The divergent half-angle was fixed at $\alpha_d = 15^\circ$ which is a widely adopted compromise for conical nozzles. Choosing a smaller divergent angle would reduce the divergence loss but at the price of a longer nozzle, increased structural mass, larger wetted area, and higher thermal exposure of the wall. Conversely, a larger angle would shorten the nozzle but would also increase the non-axial component of the exhaust velocity, thereby reducing the useful axial thrust. For this reason, $\alpha_d = 15^\circ$ was retained as a balanced solution between compactness and performance. To account for the two dimensional divergence of the exhaust flow, the corresponding correction factor was introduced, $\lambda = \frac{1+\cos \alpha}{2}$. Which, in this specific case gives $\lambda \approx 0.983$. This corresponds to a thrust loss of approximately

1.7% with respect to a perfectly axial expansion, which is sufficiently small for the purposes of the present design and remains fully acceptable within the expected uncertainty margins. The convergent half-angle was instead fixed at $\beta = 45^\circ$ to obtain a compact convergent section capable of rapidly channeling the combustion products from the chamber into the throat. A relatively steep convergent is advantageous in this case because it limits the axial extent of the upstream nozzle region, hence reducing the wall area exposed to high-temperature gases before the throat. Once the throat diameter d_t , the exit diameter d_e , and the chamber-end diameter D , were fixed by the quasi-steady ballistic sizing, the physical lengths of the divergent and convergent sections were obtained geometrically through Equation 2.6 and Equation 2.7 respectively. The total nozzle length was then evaluated in Equation 2.8.

$$L_{\text{div}} = \frac{d_e - d_t}{2 \tan \alpha} \quad (2.6) \quad L_{\text{conv}} = \frac{D - d_t}{2 \tan \beta} \quad (2.7) \quad L_{\text{noz}} = L_{\text{conv}} + L_{\text{div}} \quad (2.8)$$

The resulting final dimensions of the nozzle are summarized in Table 2.3. In this way, the nozzle design remains fully consistent with the optimized grain solution and with the real-performance reconstruction discussed in the following sections.

The calculation of the total internal lateral area is a key step for estimating the heat loads and properly sizing the cooling jacket. To improve the accuracy of the thermal analysis, the nozzle geometry is modeled as a combination of three main sections: a conical convergent, a rounded throat fillet, and a conical divergent section. Instead of approximating the nozzle with sharp transitions between cones, a rounded fillet is introduced at the throat, defined by a radius (ρ_f) and a corresponding arc. This provides a more realistic representation of the region where heat fluxes are typically highest. By separating the geometry into $A_{\text{lat,conv}}$, $A_{\text{lat,fillet}}$, and $A_{\text{lat,div}}$, the model maintains consistency across the simulation and allows a more detailed evaluation of local heat transfer effects. The results are presented in Table 2.4.

Table 2.3: Final nozzle geometry

Parameter	Value
Throat diameter, d_t [m]	0.1145
Exit diameter, d_e [m]	0.3408
Total nozzle length [m]	0.8006

Table 2.4: Internal nozzle lateral area decomposition

Section	Lateral Area [m ²]
Convergent ($A_{\text{lat,conv}}$)	0.811240
Throat fillet ($A_{\text{lat,fillet}}$)	0.011035
Divergent ($A_{\text{lat,div}}$)	0.308554
Total	1.130830

Chapter 3

Thermal Protections

Thermal protections have been designed for both the combustion chamber and the nozzle. They have been tested in order to prevent degradation due to thermal and fatigue stresses.

3.1 Case Material

The material chosen for the combustion chamber and the convergent part of the nozzle up to the area ratio of 2 is the Inconel 718. It is a material that sits at the intersection of strength, stability, and corrosion resistance, a rare combination that has made it indispensable in aerospace, energy, and advanced engineering environments. Inconel 718 belongs to a class of nickel-based superalloys that derive their strength from controlled precipitation hardening. Unlike other austenitic alloys, Inconel 718 maintains its mechanical integrity in conditions involving, such as long-term exposure to high temperatures (up to 650°C), repeated thermal cycling, high mechanical loading (tension, fatigue, and creep), and aggressive oxidizing or corrosive atmospheres. It has a low thermal conductivity that allows insulate from hot exhaust gases because heat is not transmitted quickly through the metal: in this way, by retaining heat within the gas flow, greater kinetic energy is maintained in the outgoing fluid, slightly improving thrust. On the other hand, the inner wall of the nozzle can reach temperatures above the melting point of the material itself because the heat isn't dissipated outward. However, in this analysis, the temperature at the interface between TBC and metal is considered the maximum permissible of the metal, cautiously lower than the melting temperature. The properties that have been considered are summarized in the following table 3.1:

Table 3.1: Inconel 718 properties [12, 13]

Property	Value
Density (annealed) ρ [$\frac{kg}{m^3}$]	8193.25
Conductivity k [$\frac{W}{m \cdot K}$]	27
Melting range T_{melt} [K]	1533.15-1609.15
Yield strength σ_y [MPa]	860

In addition, the thickness of the case that can withstand the maximum pressure of the system is calculated using the following formula:

$$\frac{P_{max} \cdot R}{t} = \frac{\sigma_y}{FS} \quad (3.1)$$

so $t = 6.8$ mm with $FS = 2$.

3.2 Combustion Chamber Thermal Protections

In the combustion chamber, thermal protections are fundamental to protect the metal of the case from excess heating and limit the heat radiated to associated components. Ceramic materials are a good compromise for their low thickness, which is a requested characteristics, given the reduced diameter of the grain.

Table 3.2: Comparison of thermal properties between YSZ and RSZ

Material	K_{IC} [MPa · m ^{1/2}]	k [W/(m · K)]	α [10 ⁻⁶ K ⁻¹]	E [GPa]
YSZ	2.2	0.9	10.7	30
RSZ	1.5	0.6	10.1	25

Given previous considerations and the properties reported in table 3.2, a single layer of RSZ has been chosen thanks to its low thermal conductivity, fundamental to optimize the heat exchange and a good fracture toughness, despite it being lower than the one of YSZ. The parameters needed for RSZ in the following calculations 3.3 have been computed taking a weighted average of values found in bibliography [14], [15], [16], [17] relative to its constituent materials. The principle aim of this section is to define the thickness of the

Table 3.3: RSZ thermal parameters

Property	Value
c_p [J/(kg · K)]	569.8
ρ [kg/m ³]	6601
Melting temperature [K]	2934.3

thermal protection in order to limit all possible damages to the case. Firstly, the burning time has been computed given the total impulse and the thrust generated by the SRM 3.2.

$$t_b = \frac{I_{tot}}{T} \quad (3.2)$$

The resulting burning time has indeed been set to 25 s. The system has to be operative for 29 s, considering an exceeding time of two seconds both before and after the firing. At this point, since the thickness of TBC in real applications is constant in the combustion chamber, the most critical condition has been defined to estimate the maximum required thickness. This condition is encountered at the end of the combustion chamber, since it is the point where hot gases accumulate.

3.2.1 Critical condition

In order to define the thickness of the thermal protection in the most critical condition in the combustion chamber, the equivalence of convective heat flux between the gas in the combustion chamber and the layer of RSZ and conductive heat flux through the materials

has been considered. In this condition, heat arrives to the wall of the case having a temperature that permits that it does not get damaged. The convective heat flux is defined by equation 3.3.

$$q'' = h[T_c - T_{w1}] \quad (3.3)$$

In the previous formula, T_{w1} is the temperature of the superior wall of the topcoat. Its temperature must be necessarily under the melting temperature of RSZ, $T_{melting} = 2934$ K, which is always verified. The convective heat coefficient has been evaluated considering both the Gnielinski equation 3.4 [18], which is valid for $0.5 \leq Pr \leq 2000$, $3000 \leq Re \leq 5 \times 10^6$, and the definition of Nusselt number 3.5. Values are reported in table 3.5.

Table 3.4: Temperature Values

Parameter	Value
T_{w1} [K]	2144.02
T_{int} [K]	1300
$T_{external}$ [K]	1210.33

Table 3.5: Turbulence parameters

Parameter	Value
Pr [-]	0.6088
Re [-]	1.2564×10^6

$$Nu = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (3.4) \quad Nu = \frac{hL}{k} \quad (3.5)$$

The conduction heat flux is defined by equation 3.6 and has to be evaluated for both RSZ and Inconel 718.

$$q_{cond} = k \frac{\Delta T}{d} \quad (3.6)$$

The used strategy consists in computing the temperature of the topcoat superior wall after the convective heat exchange with the gases in the combustion chamber and the thickness of the TBC by implementing an iterative algorithm B that stops when the temperature of the interface surface between RSZ and Inconel 718 is equal to the maximum temperature that the alloy can resist, increased by the previously specified safety factor. The obtained values of temperatures are reported in table 3.4. The thickness of the TBC has been defined as $d = 0.7242$ mm, which is considered to be a reasonable result. Results confirmed the optimal dissipation properties of RSZ, which is capable of reducing the temperature of 748.48K, thanks to its low thermal conductivity. In figure 3.1, this phenomenon is reported. In figure 3.2 are reported the temperature gradients of the different layers. As expected, the temperature of the hot wall of the TBC increases rapidly to a high value and gradually slows down. Indeed, the temperature increases of 1064.55 K at $0 \text{ s} < t < 1 \text{ s}$ and of 193.25 K at $1 \text{ s} < t < 2 \text{ s}$, decreasing to acceptable values after. From bibliography [2], a thermal shock has been considered to be applied increasing the temperature of 100 K/s. The results has been considered valid, despite the incongruence, due to the intrinsic nature of the problem; it has been treated making the assumption that convective and conduction heat transfers are of the same order of magnitude, as it is confirmed by the Biot number in proximity of unity. For this reason, the temperature gradient at first is greater with respect to the next seconds of firing. Indeed, considering an approach more loyal to reality, estimating with a Computational Fluid Dynamics (CFD) model the real behavior of the flow and of the heat exchange, the increase would have been more gradual. In this condition, the TBC would have resisted the thermal shock given by the combustion of the propellant, as reported in literature [2]. Moreover, the temperature reached at the

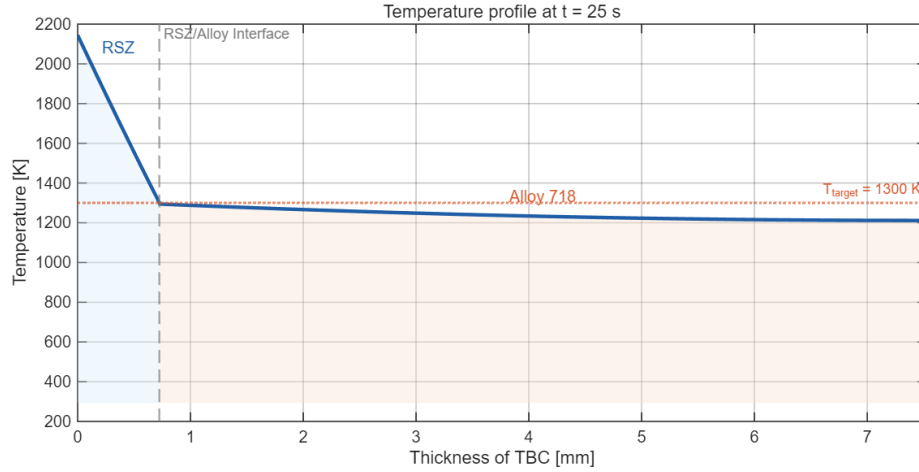


Figure 3.1: Thermal dissipation of TBC

hot wall of the RSZ layer would have been lower considering the insulating effect of the air and radiation that, for simplicity, in this report has been treated as negligible. It is also important to consider that the analyzed condition is the worst possible in the combustion chamber and that safety factors have been used. In real applications [11],

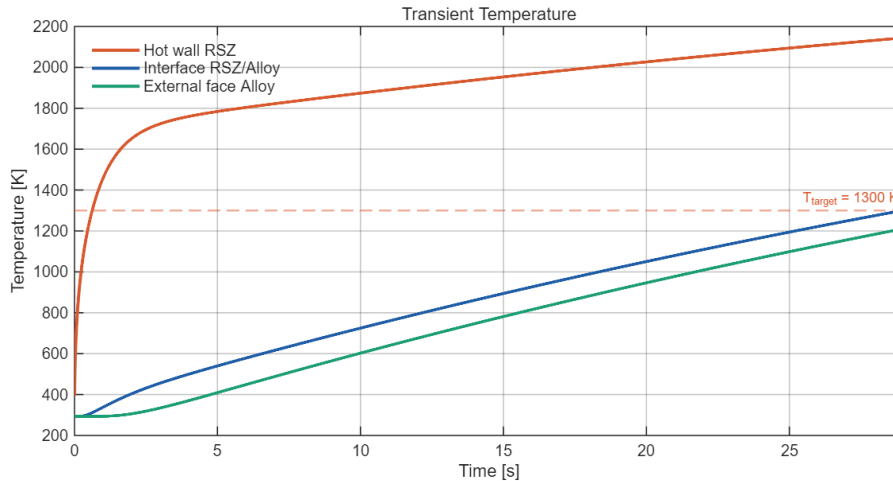


Figure 3.2: Variation of T on the surfaces

the combustion chamber is provided of a liner, which is an additional polymeric-like layer positioned between the propellant and the TBC and the TBC and the case. This material is necessary to provide adhesion and good bonding. It has been chosen to be 1.016 mm given typical applications found in literature.

3.3 Thermal Protections of the Nozzle Convergent Part

Also for the convergent part of the nozzle, thermal barrier are considered: in particular, for the part from the inlet area of the nozzle to the area ratio 2 location in the convergent part, the latter is considered as the most critical point in terms of amount of heat flux, that increases approaching the throat [19]. A study is made thinking of oversizing the system

and have no problems anywhere. Between different materials previously discussed, RSZ turns out to be the most convenient of TBCs: it has a value of thermal conductivity of $k = 0.6 \frac{W}{m \cdot K}$ and a maximum temperature to which the material does not lose its properties of 1600 K (cautiously under temperature reported in the literature) [3, 2]. If, ideally, these temperature values were not to be reached, a thinner thermal barrier thickness could be used, thus preventing common failure modes associated with ceramics subjected to extreme thermal loads, such as sintering or spallation. In this case, the temperatures are so high that TBC would not be enough to lower the interface temperature with the metal within the limits allowed by the latter without degrading (still being below the melting point). For this study, the heat flux is considered from gas to air, passing through the metal of the nozzle and TBC, using the series thermal resistance model 3.7:

$$q'' = h_g [T_{aw} - T_w] = \frac{k_{TBC}}{\delta_{TBC}} \cdot (T_w - T_{int}) = \frac{k_{met}}{\delta_{met}} \cdot (T_{int} - T_{met}) = h_{air} [T_{met} - T_{air}] \quad (3.7)$$

where h_g is the convective coefficient of gas in the nozzle, T_{aw} is the adiabatic wall temperature, T_w is the temperature of TBC interface on the gas side, k_{TBC} is the TBC conductivity, δ_{TBC} is the TBC thickness, T_{int} is the temperature of interface between TBC and nozzle metal, k_{met} is the metal conductivity, δ_{met} is the metal thickness, T_{met} is the temperature of metal on air side, h_{air} is the convective coefficient of external air and T_{air} is the air temperature. Convective coefficient of gas, h_g , is calculated considered Bartz equation [11, 20], common choice for nozzle heat transfer computation 3.8:

$$h_g = \frac{0.026}{D_t^{0.2}} \cdot \left(\frac{D_t}{R_{c,t}} \right)^{0.1} \cdot \mu_0^{0.2} \cdot \frac{c_{p,g}}{Pr_g^{0.6}} \cdot \left(\frac{p_c}{c^*} \right)^{0.8} \cdot \left(\frac{A_t}{A} \right)^{0.9} \cdot \left(\frac{\rho_{ref}}{\rho_e} \right)^{0.8} \cdot \left(\frac{\mu_{ref}}{\mu_0} \right)^{0.2}, \quad (3.8)$$

while for h_{air} is selected a value of $500 \frac{W}{m^2 \cdot K}$ [18], considering a forced convection in order to reduce the TBC thickness in the absence of a cooling jacket with water (h_w of the order of tens of thousands, so heat can be removed much faster from the external wall of the nozzle and the metal temperature dramatically decreases). In this way, the TBC thickness that can be obtained is 1.972 mm, considering an interface TBC/metal temperature of $T_{int} = 1300K$, equal to maximum temperature of Inconel 718 before melting (temperature chosen conservatively) and a conductivity of $27 \frac{W}{m \cdot K}$ [12]. It is clear that the obtained TBC thickness value is outside the normally used ranges [2, 21]: this is the result of choosing both the case material and the type of TBC. For instance, consider the case made of C/C: this material has a maximum temperature value which may be considered to be 2200 K (always cautiously below melting point [22]) and a conductivity that varies depending on the orientation of the fibers in the composite [23]. To find the thickness of the composite capable of withstanding a maximum pressure of $P_{max} = 67.5$ MPa (a $\sim 5\%$ margin above max pressure reached has been considered), the data in 3.6 were used within the formula:

$$\sigma_\theta = \frac{P_{max} \cdot D_{max}}{2 \cdot t} < \frac{\sigma_{tensile}}{FS} \quad \text{from which} \quad t > 0.029 \text{ m} \quad (3.9)$$

Keeping the same material for TBC, the values contained in 3.7 are obtained. It is noted that the values drop significantly compared to the case with Inconel 718, except in the cases with the two lowest conductivity values, which still remain out of range. Trying to change the type of TBC, switching to a multilayer of two layers with the second layer made up of RSZ and YSZ, the situation does not improve: the thicknesses increase, bringing the one to $k = 10 \frac{W}{m \cdot K}$ closer to the limit of the ranges.

Table 3.6: Characteristics of C/C

$\sigma_{tensile} = 150 \text{ MPa}$
FS = 1.5
$D_{max} = 0.8714 \text{ m}$

Even in the case of Inconel 718, changing TBC by putting the bilayer, a total thickness of 2.005 mm is obtained, still far from the ranges of the literature. This is because, adding YSZ to RSZ will increase the total thickness of TBC due to the higher conductivity of YSZ. However, on the other hand, the double layer ensures that the layer of TBC directed towards the gas (more resistant to high temperatures) does not detach from the nozzle due to the thermal expansion coefficient of the underlying layer closer to that of the metal and the lower one receives a lower temperature because it is mitigated by the external layer, keeping the properties of the materials unchanged. The adoption of the RSZ/YSZ bilayer architecture can be justified by the balance between thermal insulation and mechanical reliability, making it a more valid solution for high-temperature aerospace applications. Lastly, using the same strategy of 3.2.1 for single-layer with Inconel 718, the increase of TBC temperature in the first second of burning is $\Delta T = 1405.37 \text{ K}$, in $t = 2 \text{ s}$ it is $\Delta T = 79.65 \text{ K}$ and from there the increase gradually reduces below acceptable limits for the same reasons as cited above. The obtained thickness of TBC is 1.183 mm, less than before.

Table 3.7: Comparison of TBC thickness variation: single-layer RSZ vs bilayer RSZ/YSZ

Conductivity k [W/(m·K)]	Single-layer δ [mm]	Bilayer δ [mm]
2	3.21	3.40
5	1.46	1.55
10	0.88	0.95
11.5	0.81	0.90
21	0.58	0.65
54	0.41	0.50
80	0.38	0.45
121.5	0.35	0.45
180	0.34	0.40
233	0.33	0.40

Chapter 4

Real SRM Performance

4.1 Nominal Transient Analysis

The real performance analysis verifies how the preliminary BATES-nozzle configuration selected in `design.m` behaves under transient conditions, including time-dependent grain regression, CEA-updated thermochemistry, combustion and nozzle efficiency corrections, and robustness checks against erosive burning and parameter uncertainty. This section focuses on the workflow of `diagnosis.m`, the integrated performance metrics and the interpretation of key temporal histories, bridging nominal design targets to realistic motor operation.

4.1.1 Input Configuration from Preliminary Design

The transient analysis starts from the BATES configuration selected in `design.m`, where the preliminary sizing stage fits the Vieille’s law from strand-burner data, maps the mission requirements into average ballistic targets, and identifies the grain-nozzle layout that best balances burn time, propellant mass, mean chamber pressure and pressure flatness. The resulting configuration is summarized in section 2.4 which collect the quantities inherited by `diagnosis.m` as the nominal input state.

4.1.2 Transient Diagnosis Workflow

The script `diagnosis.m` advances the preliminary design through explicit time-stepping ($\Delta t = 0.02$ s), updating BATES geometry via radial regression, solving chamber pressure implicitly via Vieille’s law coupled to CEA-updated c^* , and computing real thrust with $\eta_{c^*} = 0.95$, $\eta_{C_f} = 0.98$. The nozzle profile is normalized for monotonic x , A/A_t , with explicit throat identification and sub/supersonic partitioning; CEA is called separately as FAC for chamber/throat equilibrium, IAC for nozzle expansion (frozen or equilibrium per freeze mode), and IAC-EQ for subsonic convergent, with results merged into a unified axial profile tagging throat, exit, and freeze point. Ignition is smoothed with $\tau_{\text{ign}} = 0.03$ s, burnout tailed off exponentially ($\tau_{\text{bo}} = 0.05$ s), yielding histories of thrust, pressure, mass flows, burning rate, c^* , I_{sp} , K_n , G_{port} .

4.1.3 Embedded FCEA2-in-the-Loop Engine

Rather than relying on a single black-box CEA evaluation, the transient solver embeds a Software-in-the-Loop (SIL) FCEA2 engine, queried via `run_cea_transient.m`, that reconstructs the motor thermodynamic state through multiple coordinated FAC and IAC calls within the time-marching loop. A FAC equilibrium calculation is first used to determine chamber and throat properties, including T_c , P_c , M_w , γ , R , c_p and c^* , while a separate IAC rocket calculation provides the nozzle-expansion quantities, such as exit Mach number, thrust coefficient and specific impulse. An additional IAC-equilibrium call is then used for the subsonic convergent branch, so that the final nozzle solution is not reduced to chamber and exit states only, but retains a spatially resolved description of the internal flow. Before these calls are executed, the nozzle station list is normalized in terms of axial coordinate and area ratio, the throat location is enforced explicitly, and the geometry is partitioned into subsonic and supersonic branches. The freezing criterion is then imposed through a dedicated configuration that allows equilibrium-to-exit expansion or frozen expansion after the chamber, the throat, or a prescribed area ratio; in the latter case, the corresponding station is inserted automatically in the nozzle discretization. The resulting CEA outputs are finally merged into a single axial profile containing x , A/A_t , T , P , Mach number, M_w , γ , C_f and I_{sp} , together with explicit identification of throat, exit and freezing point.

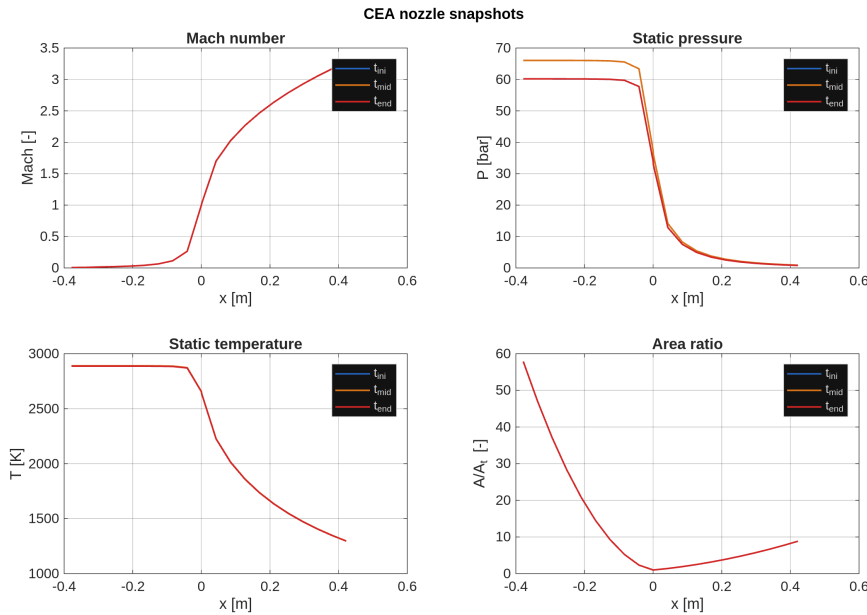


Figure 4.1: CEA-based nozzle reconstruction at representative instants of the transient. The combined FAC/IAC/IAC-EQ workflow provides spatially resolved pressure, temperature and Mach profiles, while preserving the selected freeze criterion along the divergent branch

This reconstruction is especially useful because the thrust model adopted in `diagnosis.m` is based on the corrected nozzle performance, whereas the chamber pressure iteration still depends on the theoretical c^* delivered by the equilibrium chamber solution. The algorithm therefore preserves a physically consistent separation between chamber thermochemistry, nozzle expansion and efficiency-corrected global performance, instead of embedding all effects into a single empirical correction.

4.1.4 Transient Response and Integrated Metrics

Figure 4.2 summarizes the temporal response of the selected motor configuration, showing that the transient remains well behaved throughout ignition, quasi-steady burning and tail-off. After the ignition ramp, the chamber pressure settles around the nominal operating level and stays below the 70 bar limit, while the thrust trace remains close to the 100 kN design target for most of the burning interval. The same histories also show the expected coupling between geometry and performance: as the grain regresses, K_n , mass generation rate, port mass flux and corrected nozzle performance evolve coherently with the updated burning surface and chamber state. A useful aspect of the figure is that it makes the theoretical-to-real correction immediately visible. The computed c^* and I_{sp} histories are first obtained from CEA in theoretical form and are then reduced through $\eta_{c^*} = 0.95$ and $\eta_{C_f} = 0.98$, so that the final thrust history adopted for performance assessment is the efficiency-corrected one. This explains why the motor can remain close to target thrust while exhibiting real performance levels lower than the ideal thermochemical limits, which is the expected outcome for a pre-design corrected with global combustion and nozzle efficiencies. From the integrated histories, the motor delivers a real total impulse of 2.545 MN · s, corresponding to a +1.79% deviation from the design target, with a total burn duration of 25.54 s, i.e. +2.16% relative to the nominal 25 s requirement. The average real thrust is 99.6 kN (−0.40%), the MEOP is 66.06 bar, and the total consumed propellant mass is 1092 kg. The corresponding average real specific impulse and characteristic velocity are 228.5 s and 1442 m/s, respectively, while the theoretical levels remain higher, as expected, because they do not include the efficiency penalties applied in the corrected performance model. Overall, the fused temporal and integrated view confirms that the preliminary sizing remains consistent once the full transient is introduced. The pressure response stays within the admissible envelope, the thrust level is centered on the assigned mission requirement, and the final impulse excess remains limited, indicating that the selected geometry is not only nominally feasible but also transiently robust within the adopted modeling assumptions.

4.1.5 Model Assumptions and Limitations

The transient model implemented in `diagnosis.m` adopts a quasi-steady One-Dimensional (1D) internal-ballistics formulation, with grain geometry updated through kinematic radial-axial regression and chamber pressure solved implicitly through the coupled Vieille’s law and CEA c^* iteration. Ignition and tail-off are represented through simplified exponential laws. The embedded CEA engine treats the chamber in chemical equilibrium and allows configurable freezing during nozzle expansion. Non-ideal effects are introduced through global efficiency factors, $\eta_{c^*} = 0.95$ and $\eta_{C_f} = 0.98$ (chosen as target values that would be approached in a fully refined motor), replacing a distributed loss model. Erosive burning is assessed separately and is not fed back into the main loop, while parameter uncertainties are treated through external Monte Carlo sampling rather than in-loop stochastic propagation. Wall heat transfer, ablation, and detailed boundary-layer or two-phase effects are not resolved explicitly. The model should therefore be regarded as a pre-design transient tool for performance reconstruction and trend assessment.

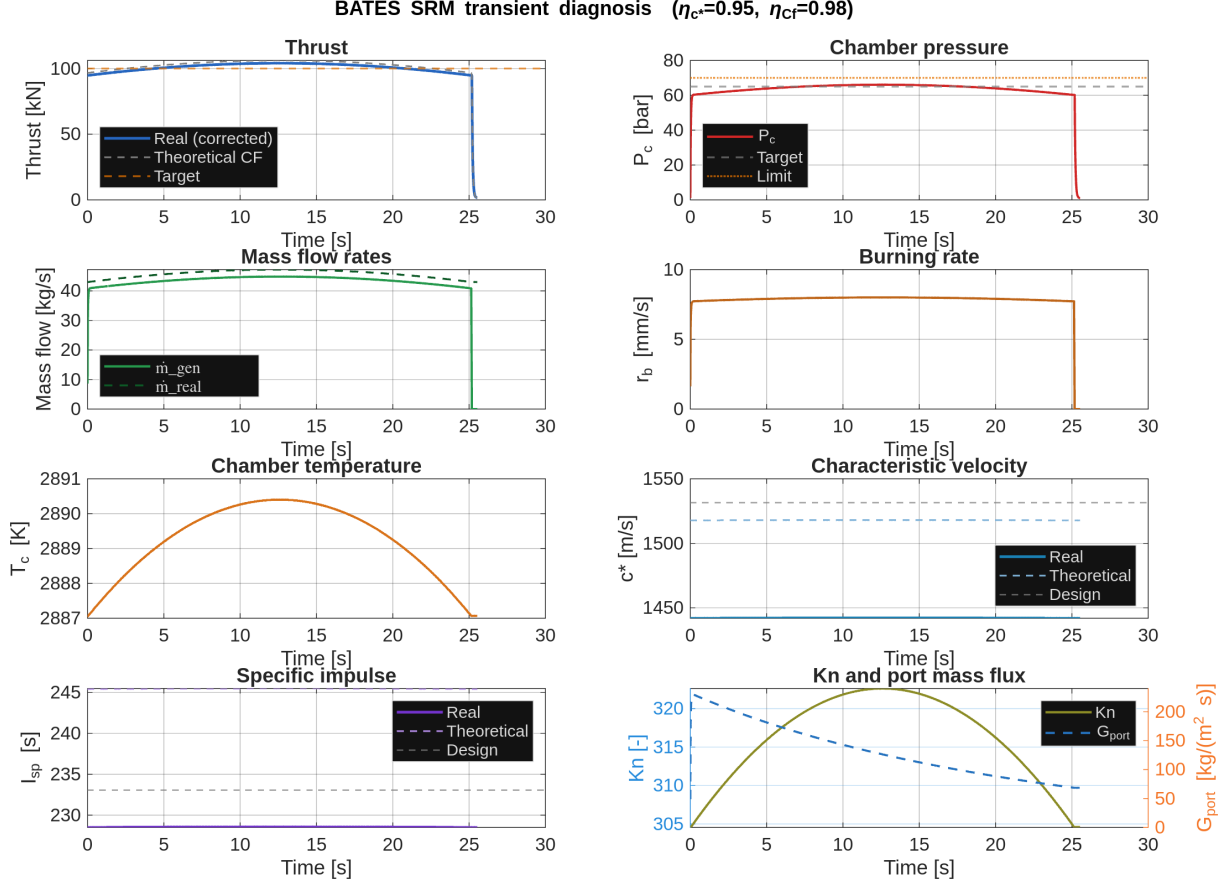


Figure 4.2: Real vs Theoretical performances - The corrected traces (solid) adopt η_{c^*} , η_{CF}

4.2 Erosive Burning Analysis

In solid rocket motors, the quasi-steady ballistic model adopted in 2.3 assumes that the local burning rate is governed solely by the chamber pressure through Vieille's law. This assumption holds when the axial flow velocity within the grain port is sufficiently low. However, when the combustion gases accelerate along the port channel, the convective heat transfer to the propellant surface is enhanced by the turbulent boundary layer, causing the local regression rate to exceed the value predicted by Vieille's law alone. Our main objective is to define whether such phenomena is present and understand the consequences it would produce on the performances. Two distinct mechanisms are recognized in the literature [24]: *velocity-based* erosive burning, driven by the core Mach number, and *mass flux-based* erosive burning, driven by the mass flux of hot gas over the burning surface G .

4.2.1 Regime study

Velocity-based criterion. For a quasi-1D isentropic flow inside the grain port, the core Mach number at the end is uniquely determined by the port-to-throat area ratio A_p/A_t through the area-Mach relation [25]:

$$\frac{A_p}{A_t} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad (4.1)$$

According to literature [24], the following thresholds are recommended for solid rocket motors:

- **Non-erosive:** $M_{\text{core}} \leq 0.50$ ($A_p/A_t \geq 1.36$ for $\gamma = 1.2$)
- **Max recommended erosivity:** $M_{\text{core}} = 0.70$ ($A_p/A_t = 1.10$)

Mass flux-based criterion: The core mass flux [24] at the end is evaluated at ignition (worst-case scenario), the non-erosive burn rate from 2.3:

$$G = \frac{\dot{m}_{\text{gas}}}{A_p} = \frac{\rho_p r_0 A_{b,0}}{A_p} \quad (4.2)$$

By linear interpolation at $p_c = 65$ bar, the non-erosive limit is $G \leq 703 \text{ kg}/(\text{m}^2 \cdot \text{s})$ and the maximum recommended erosivity is $G \approx 1757 \text{ kg}/(\text{m}^2 \cdot \text{s})$.

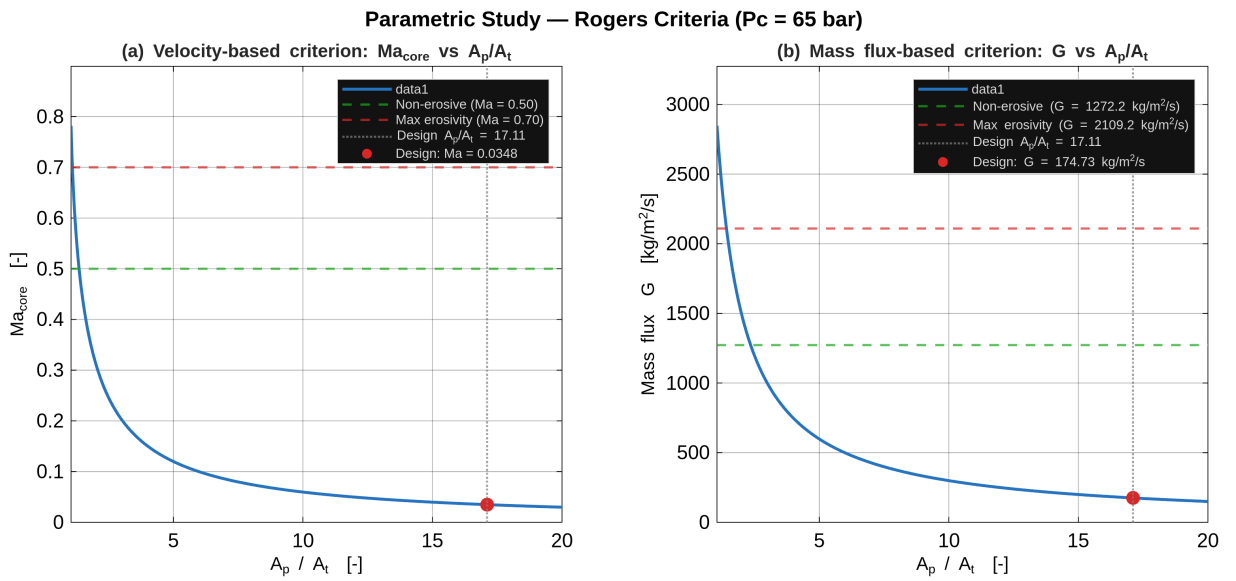


Figure 4.3: MATLAB *fzero* in-built function method used to extend the proposed graph (data 1) extracted from [24], up to the design area ratio

Table 4.1 and Figure 4.3 summarize the erosive burning diagnostic for the chosen grain. The port-to-throat area ratio has been computed knowing the diameters, giving $A_p/A_t = 17.11$, far above the non-erosive threshold of 1.36. Numerical inversion of Eq. (4.1) with $\gamma = 1.164$ is not directly possible but requires numerical methods. As a starting point, a table has been used [24] considering a $\gamma = 1.2$, allowing to directly obtain the Mach number through an *fzero* function. Both criteria confirm that the motor operates deeply within the non-erosive regime, thus a standard Vieille-based ballistic model is fully adequate for performance prediction.

Table 4.1: Erosive burning diagnostic summary at ignition

Parameter	Design value	Non-erosive threshold
A_p/A_t [-]	17.11	≥ 1.36
M_{core} [-]	0.035	≤ 0.50
G [$\text{kg}/(\text{m}^2 \cdot \text{s})$]	174.73	≤ 703

4.2.2 Evaluation on performances and critical assessments

Since this phase of the project considers a groundwork approach to the Stark-100k, the team decided to include a preliminary study of such physical phenomena on the main performances. The total propellant burning rate can be decomposed as $r_b = r_0 + r_e$, where r_0 is the non-erosive (Vieille) contribution and r_e is the erosive augmentation. Distinct methodologies have been evaluated, differing primarily in the physical parameter chosen as the correlation variable. Simpler approaches relate the burn rate augmentation to the centerline Mach number or mass flux, yielding models that are straightforward to implement in 1D ballistic codes but fail to scale correctly to large motors [26]. More advanced semi-empirical models adopt dimensionless parameters based on normalized mass flux and a local Reynolds number, demonstrating universal validity across propellant types and grain geometries without propellant-specific calibration [27, 28]. The most physically rigorous strategies would require CFD simulations, an approach deemed not worth time and cost during this phase of the project [26]. A BATES motor features cylindrical grains burning both radially and axially, designed to produce a nearly constant thrust profile. Erosive burning significantly disrupts this balance. At ignition, the port area would be realistically minimum and the mass flux is maximum, generating a burn rate peak at the end of the grain, allowing for over-pressurization at ignition thus producing a thrust higher than the nominal value followed by a rapid decay [24]. The accelerated regression may also prematurely expose the thermal insulation, and the MEOP is systematically underestimated when erosive burning is neglected in the ballistic model. To counteract these effects, the Rogers requirements should be respected, for instance, by keeping a constant core mass flux design (achieved by progressively increasing the port diameter across successive grains), maximizing propellant loading without increasing erosivity [24].

4.3 Monte Carlo Analysis

To evaluate the robustness of the selected BATES motor design under parameter uncertainty, a Monte Carlo analysis was carried out using the updated transient internal-ballistics workflow. In the current implementation, each realization is propagated through the same quasi-steady 1D solver adopted for the performance diagnosis, so that the statistical outputs remain consistent with the updated analysis chain rather than with a purely idealized ballistic reconstruction. The uncertain inputs were selected to represent the dominant sources of variability affecting the ballistic response of the motor. In particular, the Vieille's law parameters a and n , were treated as uncertain because they were obtained from the regression of experimental ballistic data and therefore carry statistical estimation uncertainty. In addition, the propellant density $\rho_{propellant}$ and the characteristic velocity c^* were also perturbed in order to account for residual variability in propellant loading and effective thermochemical performance. The total number of Monte Carlo realizations was fixed at $N_{mc} = 2000$, which represents a reasonable compromise between computational cost and statistical convergence. In addition to the final distributions, convergence was monitored through the running mean and running standard deviation of MEOP. For each realization, the uncertain parameters were sample as

$$a_k = a + \sigma_a \mathcal{N}(0, 1) \quad (4.3) \quad n_k = n + \sigma_n \mathcal{N}(0, 1) \quad (4.4) \quad \rho_{p,k} = \rho_p + \sigma_\rho \mathcal{N}(0, 1), \quad (4.5)$$

and, in the extended sensitivity formulation implemented

$$c_k^* = c^* + \sigma_{c^*} \mathcal{N}(0, 1), \quad (4.6)$$

where $\mathcal{N}(0, 1)$ denotes a standard normal random variable. The sampled values were then propagated through the time-marching internal-ballistics model, yielding a population of output realizations for MEOP and burning time.

4.3.1 Output metrics

The Monte Carlo simulation results are presented in Table 4.2, comparing the mean values against the design targets.

Table 4.2: Monte Carlo results for ballistic-performance uncertainty

Output quantity	Mean, μ	Standard deviation, σ
MEOP [bar]	62.06	6.29
Burning time [s]	25.91	2.55
Total impulse [N s]	2.380564×10^6	3.3620×10^4

4.3.2 Analysis of Pressure (MEOP), Total Impulse and Burn Time

The simulated MEOP obtained is 62.06 bar, slightly below the 65 bar starting point, corresponding to a 4.52% deviation. Although acceptable at pre-design level, this suggests adopting conservative structural margins. Using a 3σ upper bound, the chamber pressure may reach 80.93 bar, which should therefore be regarded as the minimum burst-pressure requirement for the casing. The standard deviation of 6.29 bar indicates moderate sensitivity to uncertainties in the Vieille’s law parameters and should be accounted for in both structural sizing and cooling-jacket design. The motor remains highly consistent in terms of total impulse, which stays essentially on target with a coefficient of variation of 1.41%. This confirms that burn-rate variations have limited impact on the total energy released, since the propellant mass is fixed. Burn time, however, shows a wider spread of ± 2.55 s, which directly affects thrust delivery: shorter burns correspond to higher thrust levels. In this way, while total energy remains stable, its temporal distribution does not, with direct implications for load history and system-level structural assessment. The histograms in Figure C.1 show that both MEOP and burn time remain centered around well-defined mean values, with no evident tails or secondary peaks. The convergence plots further support the reliability of the statistical analysis: the mean MEOP stabilizes well before the end of the 2000 samples, with only minor fluctuations after roughly 500 iterations, and the standard deviation reaches a clear plateau (stability). This indicates that the adopted sample size is sufficient to resolve both the mean behavior and the associated spread. The main limitation of the present analysis is that thermochemical performance is represented through a statistical perturbation of the characteristic velocity, without explicitly modeling its dependence on the instantaneous chamber state.

Chapter 5

Cooling Jacket & Tank design

5.1 Nozzle Material

The selected nozzle material is CuCrZr, a precipitation-hardened copper alloy widely used in regenerative cooled rocket engines such as Vulcain and RS-68. It is chosen primarily for its high thermal conductivity ($\sim 320 \text{ W}/(\text{m} \cdot \text{K})$) [29], which ensures efficient heat transfer from the hot gas wall to the coolant, helping to maintain manageable wall temperatures under extreme heat fluxes. CuCrZr derives its strength from Cr_2Zr precipitates formed during heat treatment. However, this strengthening mechanism is temperature sensitive: above 400°C ($\sim 673 \text{ K}$), precipitates dissolve, leading to a significant loss of mechanical strength. For this reason, a conservative maximum service temperature of 300°C ($\sim 573 \text{ K}$) is adopted to preserve mechanical integrity under cyclic thermal loading. The properties considered for the nozzle wall are the following:

Parameter	Symbol	Value
Wall thickness	t_{wall}	$2.19 \times 10^{-3} \text{ m}$
Thermal conductivity (CuCrZr)	k_{wall}	$320 \text{ W m}^{-1} \text{ K}^{-1}$
Maximum wall temperature	$T_{\text{wall,max}}$	573 K

Table 5.1: Wall material properties and thermal limits

5.2 Need of a Thermal Barrier Coating

5.2.1 Gas/Wall, Convective Heat Transfer Coefficient

Using the gas mass flow rate $\dot{m}_{\text{gas}}$, and the variation of the cross section between the convergent/throat and the divergent/throat areas ratio, both equal to 2, it is possible to recover the convective heat transfer coefficient determination h_g using the Dittus-Boelter correlation. Although the Prandtl number is close to the lower limit of the validity range ($\text{Pr} = 0.6088$), the Dittus-Boelter correlation remains reasonably applicable because the flow is fully turbulent ($\text{Re} = 53234$). Nonetheless, Bartz's model was also implemented in order to have an element of comparison and therefore validate the model.

5.2.2 Adiabatic wall temperature

Knowing the specific heat ratio γ , it is possible to recompute the Mach number of the flow all over the domain cross section area using formula 4.1; the figure can be found in the appendix C. Consequently, it is possible to observe that between the two area ratio, given $M > 0.3$, the flow is therefore compressible and the adiabatic wall temperature can be computed using the recovery factor, as such $T_{aw} = RT_{cc}$.

5.2.3 Interface metal/gas temperature determination

In order to assess the necessity of a TBC, a preliminary thermal analysis is performed considering only the gas-side convection and the conductive resistance of the metallic wall. Knowing h_g via the Dittus-Boetler correlation, R_{gas} can be computed and so the total thermal resistance without TBC, $R_{s,TBCs} = R_{gas} + R_{wall}$. The corresponding heat flux is then obtained and therefore, the hot wall temperature (gas-side wall temperature):

$$q = \frac{T_{aw} - T_{bulk}}{R_{s,TBC}} \quad T_{w,h} = T_{aw} - q R_{gas} \quad (5.1)$$

It has to be noted that the conductive thermal resistance of the wall is evaluated using three different modeling approaches, two simplified 1D formulation assuming a planar geometry (using Dittus-Boetler/Bartz) and a more accurate model based on cylindrical geometry (using Dittus-Boetler). These models will also be compared in the following parts of the TBC and cooling jacket design. It can be observed in the following table, that, as expected, the most constraining conditions are situated at the throat level ($x = 0$), but also that $T_{interface}$ is much higher in the models than $T_{wall,max} = \frac{573}{1.2} = 477.5$ K (figure in Appendix E), which highlights the need to add a TBC whose thickness will be computed in the next parts.

	Plane model	Cylindrical model	Bartz model
$T_{max, interface}$ [K]	601.3956	596.5565	408.2897

Table 5.2: Maximum gas-side interface temperature for different heat transfer models

5.2.4 Properties of the TBC

The TBC selected for this study is yttria-stabilized zirconia (YSZ), specifically 8 wt% yttria-stabilized zirconia due to its low thermal conductivity, high temperature resistance, and good compatibility with metallic substrates. Its microstructure, particularly when deposited using APS, leads to a porous structure that significantly reduces thermal conductivity and enhances its insulating performance. The thermal conductivity of YSZ typically ranges between 0.9 and 2.5 W/(m · K) depending on the porosity level and deposition process [1]. In this work, a conservative value of $k_{TBC} = 0.9$ W/(m · K) is adopted in order to maximize the thermal insulation effect and ensure a safe estimation of the wall temperature.

5.3 Tradeoff TBC / Cooling jacket design

5.3.1 Cooling jacket geometry

The cooling system is designed as a co-flow arrangement consisting of multiple parallel cooling tubes distributed along the chamber wall. Each tube has an inner diameter of 2 mm, selected as a compromise between manufacturability, hydraulic performance, and the ability to achieve high convective heat transfer coefficients through sufficiently large coolant velocities. The total number of tubes is not fixed a priori and is treated as a key design variable within the sizing procedure. It is determined through an iterative process that will be described in the next sections. The coolant pressure is set to 9.75 bar, corresponding to approximately 15% of the chamber pressure p_c [30]. At this operating pressure, the saturation temperature is approximately $T_{sat} \approx 455$ K, which defines the upper thermal safety limit to avoid phase change within the cooling channels. The coolant operating conditions are summarized in the following table:

Parameter	Value
Configuration	Co-flow cooling jacket (parallel tubes)
Tube inner diameter	2 mm
Coolant pressure P	9.75 bar ($\approx 0.15 P_c$)
Saturation temperature T_{sat}	455 K
Inlet temperature T_{in}	291.15 K
Outlet temperature T_{out}	323.15 K

Table 5.3: Cooling jacket geometry and operating conditions

5.3.2 Minimal thickness of TBCs

From the literature [1], it is known that typically the thickness of the TBC ranges up to 1 mm. Therefore, an interval of thickness of up to 1 mm μm is considered. Due to the low coating thickness, the planar conduction model is adopted for the TBC thermal resistance, hence the total thermal resistance defined as $R_{tot} = R_{gas} + R_{wall} + \frac{d}{k_{TBC}}$. The resulting heat flux is then determined and the hot-side wall temperature (at the interface between the *CuCrZr* and the TBC) is obtained by subtracting the temperature drop across the gas and TBC resistances:

$$q = \frac{T_{aw} - T_{bulk}}{R_{Tot}} \quad T_{w,h} = T_{aw,max} - q(R_{gas} + R_{TBC}) \quad (5.2)$$

An iterative procedure is then performed over the range of TBC thickness values in order to identify a configuration that satisfies the thermal constraint $T_{w,h} < \frac{T_{wall,max}}{1.2}$. The safety factor of 1.2 is introduced to ensure the conservation of the metal *CuCrZr*. The results indicate that the minimum TBC thickness required, for both planar (39 μm and 20 μm) and cylindrical (37 μm) formulations, falls within a physically reasonable range. Once the optimal TBC thickness is defined, the heat flux to be removed by the cooling system is fully determined. Assuming steady-state conditions, the heat flux is conserved across the system, such that the heat transferred from the gas side equals the heat removed by the coolant. The required convective heat transfer coefficient is then evaluated (with a

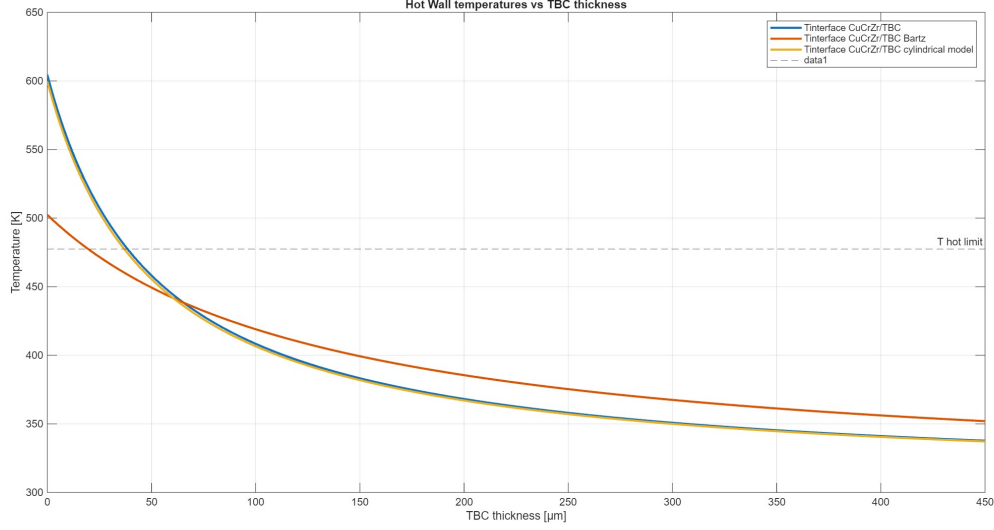


Figure 5.1: Evolution of $T_{interface,TBC}$ with the TBC thickness

safety factor of 1.2 to ensure that $T_{wall,cold} < T_{sat}$), and the corresponding coolant velocity is obtained using the Dittus–Boelter correlation.

$$h_{req} = \frac{q}{\left(\frac{T_{sat}}{1.2} - T_{bulk,w}\right)} \quad v_{req} = \frac{Re \mu_w}{\rho_w D} \quad (5.3)$$

5.3.3 Required coolant velocity

The coolant velocity is a key design parameter, as it directly governs the convective heat transfer coefficient and therefore the overall cooling performance. In typical cooling systems, literature values generally range between 5 and 15 m/s. Below this range, heat transfer becomes insufficient, while higher velocities lead to excessive pressure drops and increased pumping requirements. In the present configuration, using the previously determined minimum TBC thickness, a required coolant velocity of $v_{req} = 57.35$ m/s is obtained. This value significantly exceeds the acceptable range, indicating that the cooling demand is too severe for the current design. This highlights a fundamental trade-off between thermal insulation and hydraulic performance: increasing the TBC thickness reduces the heat flux and therefore decreases the required coolant velocity, but excessive insulation may introduce mechanical and integration limitations. An iterative design procedure is therefore adopted to ensure both constraints are satisfied ($v_{req} < 15$ m/s and $d < 350$ μm), leading to a balanced and feasible configuration. It can be observe that for both model (plane/cylindrical) a thickness of respectively 349 μm (270 μm using Bartz model for h_g) and 347 μm (figure in Appendix D) is enough to ensure that v_{req} stays below the reasonable threshold fixed previously.

5.3.4 Coolant real velocity v_{real}

Once the optimal TBC thickness is determined, the corresponding heat flux entering the cooling system is fixed. Under steady-state conditions, the heat removed by the coolant

is assumed equal to the heat received from the hot gas:

$$Q = \dot{m} c_p (T_{out} - T_{in}) \quad (5.4)$$

which directly defines the total coolant mass flow rate $\dot{m}$. The cooling jacket consists of N identical parallel tubes of inner diameter D , as a consequence, the coolant velocity inside each tube is then given by:

$$v_{real} = \frac{\dot{m}}{\rho_w N A} \quad (5.5)$$

Since the required velocity v_{req} is determined from the heat transfer requirement imposed by the TBC-controlled heat flux, the number of tubes N is obtained through an iterative procedure. A discrete range of possible tube numbers is tested, and the corresponding velocity is evaluated at each step. The optimal configuration is selected when the relative difference between the actual and required velocities satisfies:

$$\frac{|v_{real} - v_{req}|}{v_{req}} < 0.01 \quad (5.6)$$

Thus, it ensures a feasible design that satisfies both thermal requirements (set by the TBC-driven heat flux) and hydraulic constraints (maximum allowable coolant velocity). Following this computation it appears that 50 tubes were necessary for both models.

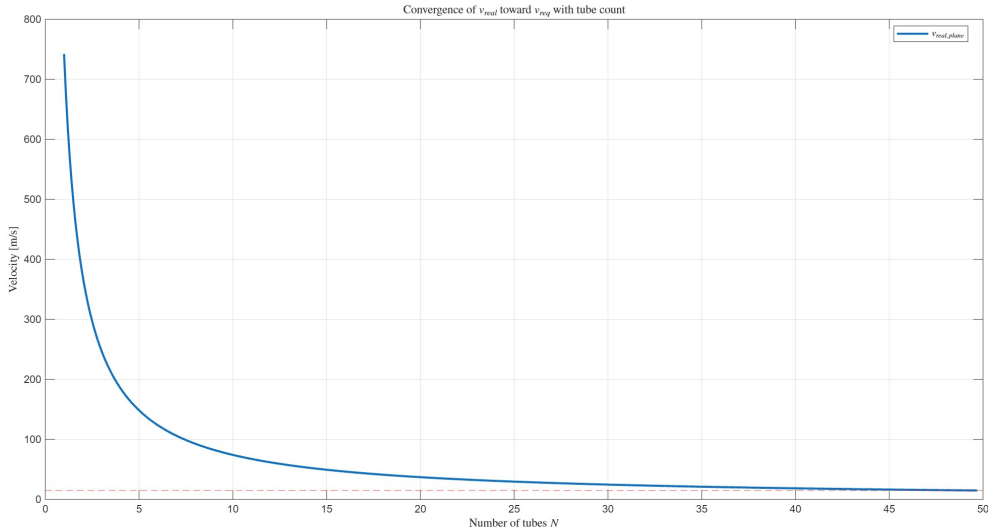


Figure 5.2: Convergence of v_{real} towards v_{req} with tube count

5.3.5 Design validation

The results show that, in all three thermal models (plane, cylindrical, and Bartz), the actual coolant velocity slightly exceeds the required velocity. This ensures that the cooling system provides a sufficient safety margin in terms of convective heat removal capacity. In addition, the interface temperature between the metal and the coolant remains significantly below the coolant saturation temperature T_{sat} . This indicates that no phase change

or boiling phenomena occur within the cooling channels, guaranteeing stable single-phase liquid cooling conditions throughout the system. As a result, the risk of thermal instabilities such as local vapor formation, flow choking, or critical heat transfer deterioration is avoided, ensuring safe and reliable operation of the cooling jacket. In all three models,

Model	$T_{\text{interface,coolant}}$ [K]	v_{req} [m/s]	v_{real} [m/s]	$T_{\text{interface,TBC,max}}$ [K]
Plane model	378.7782	14.8033	14.9361	412.0855
Cylindrical model	378.7389	14.9020	15.0460	411.5587
Bartz model	378.8078	14.8416	14.9670	412.6110

Table 5.4: Comparison of thermal and hydraulic results for different heat transfer models

the introduction of the TBC ensures that the metal/TBC interface temperature remains below the allowable limit $T_{\text{max}} = 573$ K along the entire axial profile. Also, a maximum temperature gradient of about 34 K is observed across the wall (378 K on the coolant side and 412 K on the TBC side on plane model), inducing thermally generated stresses. An estimation using $\sigma = E\alpha\Delta T$ gives $\sigma \approx 63.6$ MPa, which is well below the yield strength of CuCrZr (200–300 MPa [29]). This indicates that the thermal stresses are moderate and compatible with cyclic operation.

5.3.6 Status of TBCs

The temperature at the gas–TBC interface is significantly higher than recommended operating conditions, which may promote degradation mechanisms such as sintering, phase transformations, and loss of strain compliance [3]. Furthermore, accelerated growth of the thermally grown oxide (Thermally Grown Oxide (TGO)), known as Rapid Growth of Oxide (RGO), can induce additional stresses due to thermal expansion mismatch, potentially leading to cracking or delamination. However, the temperature remains below the melting point of YSZ ($\sim 2700^\circ\text{C}$ [31]), so no immediate melting is expected. It would nevertheless be relevant to assess the time-dependent degradation of the TBC over the 29 s burn duration, as well as its impact on the underlying metal.

5.4 Tank sizing

From the cooling jacket design, using the plane model, it is clear that the mass flow rate of the water flowing inside the cooling jacket is $\dot{m}_{\text{water}} = 2.3041$ kg/s and the heat load is $Q = 3.0824 \times 10^5$ W. In a steady state flow, the heat removed by the water must balance the heat produced by the engine.

$$\Delta T = \frac{Q}{\dot{m} \cdot c_p} \approx 32^\circ\text{C}, \quad (5.7)$$

where $c_p = 4186$ J/(kg · K). Given that $T_{\text{in}} = 18^\circ\text{C}$, $T_{\text{out}} = 50^\circ\text{C}$ and this is a safe value with respect to the boiling point that at a pressure of 9.75 bar is $T_{\text{boiling}} = 178^\circ\text{C}$. For the hydraulic design of external lines, it was assumed $v = 2.2$ m/s [32] to reduce vibrations and losses. Considering that the density of water is $\rho = 1000$ kg/m³, we can calculate the diameter of the pipe:

$$D = \sqrt{\frac{4\dot{m}}{\pi\rho v}} = 36.5 \text{ mm} \quad (5.8)$$

So, the most suitable commercial choice is the DN40 stainless steel line, made of AISI 316L [33, 34, 35].

The evaluation of the pressure losses, given that it has been assumed that the flow is turbulent, can be modeled using Darcy-Weisbach formulation.

$$Re = \frac{\rho v D}{\mu} \approx 8 \times 10^4 \quad (5.9) \quad f = 0.316 Re^{-0.25} \approx 0.02 \quad (5.10) \quad \Delta P_{dist} = f \frac{L}{D} \frac{\rho v^2}{2} \approx 0.13 \text{ bar}, \quad (5.11)$$

where $\mu = 1 \times 10^{-3} \text{ Pa} \cdot \text{s}$, and as $Re \gg 4000$ the flow is considered to be turbulent. This is consistent with the adopted hydraulic model and is beneficial to heat transfer, although it increases pressure losses. As the length of the pipe, it is assumed that $L = 10 \text{ m}$. This choice was made as the length is determined by the physical layout of the experimental rig including the tank, the valves, the filter, the piping, the cooling jacket and the discharge point.

$$\Delta P_{conc} = \sum K \cdot \frac{\rho v^2}{2} \approx 0.21 \text{ bar} \quad (5.12)$$

Local losses are due to the components of the experimental circuit. For a preliminary sizing, it was assumed the presence of an inlet/outlet of the line, bends, Y-strainer, check valve and a possible quick shut-off valve. For the two 45° bends $K_{curve} = 2 \times 0.4 = 0.8$, for the Y-strainer $K_{strainer} = 2.0$, for the check valve spring-loaded $K_{check} = 2.5$, for the pneumatic quick-opening valve $K_{valve} = 0.2$, for the inlet/outlet of the line $K_{in/out} = 1.5$, and finally for the inlet/outlet manifolds it was considered $K_{manlet} = 1.5$ so, in total, $\sum K = 8.5$ [36]. This is a realistic estimate for a compact experimental loop. If the check valve introduces significant losses or the filter is too restrictive, this value can increase substantially.

$$\Delta P_{tot,ext} = \Delta P_{dist} + \Delta P_{conc} \approx 0.34 \text{ bar} \quad (5.13)$$

It has been decided to approximate $\Delta P_{tot,ext} = 1.0 \text{ bar}$ to have a conservative estimate for experimental design and it ensures a sufficient pressure margin. For the final pressure, it has been considered the required pressure in the cooling jacket ($P_{j,in} = 9.75 \text{ bar}$) and the total pressure lost ($\Delta P_{tot,ext} = 1.0 \text{ bar}$):

$$P_{fin} = P_{j,in} + \Delta P_{tot,ext} = 10.75 \text{ bar} \quad (5.14)$$

To limit the pressure decay during blowdown, the initial gas pressure is selected as 1.4 times the final gas pressure. Since gas expansion laws require absolute pressure, the pressure ratio is applied in absolute units.

$$P_{fin,abs} = 10.75 + 1 = 11.75 \text{ bar} \quad (5.15) \quad P_{in,abs} = 1.4 P_{fin,abs} = 16.45 \text{ bar} \quad (5.16)$$

For the blowdown tank design, it is necessary to consider that the system must operate continuously for 29 s.

$$V_{H_2O} = \frac{\dot{m}}{\rho} \cdot t_{tot} \cdot FS \approx 0.0836 \text{ m}^3 \quad (5.17)$$

As the safety factor, it was chosen $FS = 1.25$ as it prevents gas ingestion into the jacket and it accounts for uncertainties. To determine the gas volume, it was assumed a polytropic gas expansion:

$$V_{gas} = \frac{V_{H_2O}}{\left(\frac{P_{in}}{P_{fin}}\right)^{1/n_p} - 1} \approx 0.283 \text{ m}^3 \quad (5.18)$$

As the polytropic exponent it was chosen $n_p = 1.3$ for N_2 . It is a suitable choice as it represents fast expansion conditions.

$$V_{tot} = V_{H_2O} + V_{gas} \approx 0.367 \text{ m}^3 \quad (5.19)$$

The theoretical minimum total tank volume is approximately 0.367 m^3 . A 350 L tank would be insufficient, whereas a 400 L tank provides the first suitable commercial size above the required value, with a moderate design margin of about 9%. For this reason, a 400 L stainless steel tank is considered the most appropriate practical choice.

A preliminary tank wall thickness can be estimated using thin-walled cylindrical pressure vessel theory [37], since the expected thickness remains much smaller than the vessel diameter. The pressure used for the calculations is the design pressure $P_{design} = 1.5 \cdot P_{op} = 23.17 \text{ bar}$ where the $P_{op} = 15.45 \text{ bar}$ is the highest internal gauge pressure in the tank during operation ($P_{op} = P_{in,abs} - 1$). Stainless-steel has an allowable stress in the range 115-130 MPa and we can assume a tank diameter in the range 0.5-0.7 m.

$$t = \frac{pD}{2\sigma_{amm}E} \approx 0.00830 \text{ m} \quad (5.20)$$

The pressure was converted in MPa, so $p = 2.317 \text{ MPa}$, for the diameter was chosen $D = 0.70 \text{ m}$, for the stress it was chosen $\sigma_{amm} = 115 \text{ MPa}$ and as weld efficiency was chosen $E = 0.85$.

The system includes a $100 \text{ }\mu\text{m}$ Y-filter to prevent the clogging of the cooling channels, a check valve to avoid backflow and gas ingress and a pneumatic valve to control the timing of the coolant flow. A burst disk provides overpressure protection, while a nitrogen pressurization system is used before the test to pressurize the tank, ensuring autonomous operation without external supply. The external pipeline is not connected directly to each cooling channel, but through an inlet manifold that distributes the total coolant flow to the 50 parallel channels of the cooling jacket. A corresponding outlet manifold then collects the flow and merges it again into a single discharge line. [38]

Thermocouples should be provided at the cooling water inlet and outlet and at the nozzle/cooling jacket assembly outer wall to monitor system thermal response during the test. Thermocouples are used to verify calculated coolant temperature rise, record structural temperature and identify potential local hot spots if they exist. Thermocouples were chosen for this application because they are simple, robust and can operate in severe thermal environments. It should be noted that wall thermocouples measurements provide the temperature of the external surface and not directly the internal hot-gas-side wall temperature[39, 40].

Chapter 6

Conclusions

This work developed a pre-design framework for the Stark-100k motor that combines propulsion, thermal, tank, and robustness analyses within a single workflow. The ballistic characterization provided the pressure-law closure required to connect experimental regression data with motor-scale performance predictions, while the thermochemical screening supported the selection of an AP/HTPB formulation that balances delivered performance with acceptable flame temperature and thermal loading. On this basis, the BATES grain and nozzle geometry were sized to satisfy thrust, total impulse, burn-time, and pressure-flatness requirements, while preserving a configuration suitable for stable heat-transfer investigation at the throat. The subsequent transient analysis confirmed that the selected layout remains close to the design targets once geometry evolution, updated CEA properties, and non-ideal performance corrections are introduced. The erosive-burning study further supported this result by showing that the nominal design operates deeply within the non-erosive regime, thereby justifying the use of the adopted pressure-based ballistic model for the present configuration. The Monte Carlo analysis added a robustness perspective, indicating that the configuration is not only nominally feasible but also reasonably stable with respect to the dominant ballistic and thermochemical uncertainties. A major outcome of the work also concerns the thermal-management architecture. The report showed that the throat experiment, the reusable cooling-jacket concept, the chamber thermal-protection strategy, and the associated tank subsystem cannot be decoupled from the propulsion design, because combustion temperature, heat flux, coolant inventory, residence time, and geometric stability directly affect both thermal boundary conditions and hardware survivability. In this context, the RSZ-based protection concept, the jacket sizing, and the preliminary tank definition were treated as design variables linked to allowable wall temperature, repeatability of the test condition, and feasibility of repeated firings, rather than as secondary auxiliaries. At the same time, the present framework retains the typical limitations of a pre-design tool. The flow is quasi-steady and one-dimensional, the cooling loop and tank subsystem are addressed only at preliminary level, and the thermal protection is sized without a fully coupled degradation model including conjugate heat transfer, coating damage, and structural response. Future work should therefore proceed along three main directions: a coupled thermo-fluid model of chamber, throat, jacket, and tank-fed cooling system; a transient thermo-mechanical assessment of the TBC including cracking, sintering, and durability; and an experimental calibration of ballistic, efficiency, and thermal-loss models through static-fire and thermal-loop data. With these extensions, the present methodology could evolve from a coherent pre-design tool into a predictive framework for test planning, hardware refinement, and progressive qualification of the Stark-100k system.

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Appendix A

Design and Performance in depth

For each pair (N, w) , the diameter of the outer grain was not arbitrarily selected, but was obtained algebraically by enforcing consistency with the target average burning area (recall it is retrieved from the NASA CEA output). In this way, each tested geometry already satisfies the required order of magnitude for mass flow rate and thrust, and the optimization focuses only on corrections associated with the pressure trace shape and burn duration accuracy. Once D was obtained, the initial port diameter was evaluated from the web definition in Equation 2.2 and the segment length was assigned through Equation 2.3 so that the peak burning area occurs extremely close to mid-burn, promoting a more symmetric pressure evolution between ignition and burn out.

Only physically admissible configurations were retained. In particular, cases with an excessively small initial port diameter or non-meaningful segment lengths were discarded before the ballistic reconstruction step. In this sense, the remaining candidates were then evaluated through a quasi-steady regression simulation performed over the regression coordinate x , from $x = 0$ to $x = w$.

At each step, the instantaneous port diameter and segment length were updated according to Equation A.1 and Equation A.2

$$d(x) = d_0 + 2x \quad (\text{A.1}) \quad L(x) = L_0 - 2x \quad (\text{A.2})$$

and the corresponding burning area was computed as

$$A_b(x) = N\pi \left[d(x)L(x) + \frac{1}{2} (D^2 - d(x)^2) \right]. \quad (\text{A.3})$$

The evolution of the geometric parameter K_n was then obtained as

$$K_n(x) = \frac{A_b(x)}{A_t} \quad (\text{A.4})$$

By substituting Vieille's Law into the mass balance equation ($\rho_p A_b r_b = P_c A_t / C^*$), the code solves for the equilibrium chamber pressure. The resulting formula, implemented in the optimization loop to evaluate the pressure in bar, is:

$$P_{c,\text{bar}}(x) = \left(\frac{K_n(x), a, \rho_{\text{propellant}}, C^*}{10^8} \right)^{\frac{1}{1-n}} \quad (\text{A.5})$$

The term 10^8 in the denominator acts as a scaling factor to maintain consistency between the SI units of density and C^* and the pressure expressed in bar. This pressure value is

then used to update the local burning rate $r_b(x)$, allowing for the calculation of the total burn time t_b through numerical integration along the regression path.

On the other hand, Figure A.1 presents a zoomed-in plot of Figure 4.2, where k_n and the chamber pressure can be observed more in detail.

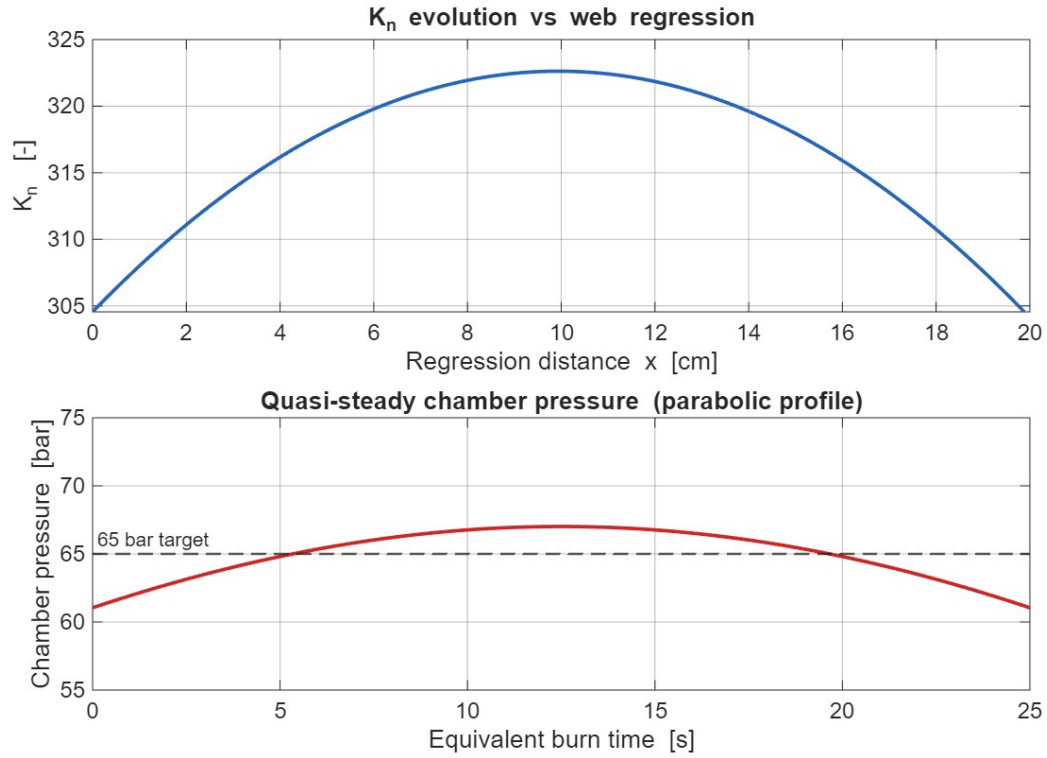


Figure A.1: In depth behavior of K_n and Chamber Pressure.

Appendix B

Combustion Chamber Algorithm

The algorithm implemented in Matlab to compute the thickness of the TBC in the combustion chamber uses an iterative process to find the unknowns. The aim of the script is to define the needed thickness in order to guarantee the integrity of the metal case, defining also the temperature reached at the superior wall of the RSZ layer during the convective heat exchange with the hot gases.

The algorithm uses a finite difference method with an explicit scheme to solve the heat equation.

Since the system is not treatable as a continuum from the computation point of view, the domain has been discretized in multiple nodes.

At nodes describing the same material, thermal and physical properties are constant. Indeed, it is possible to use the general discretized conduction equation. It has been discretized using forward Euler discretization in time and centered differences in space. Equations B.1 and B.2 show respectively the equations for RSZ and Inconel 718.

$$T_i^{n+1} = T_i^n + \frac{\alpha_{RSZ}\Delta t}{\Delta x_{RSZ}^2}(T_{i+1}^n - 2T_i^n + T_{i-1}^n) \quad (\text{B.1})$$

$$T_i^{n+1} = T_i^n + \frac{\alpha_{alloy}\Delta t}{\Delta x_{alloy}^2}(T_{i+1}^n - 2T_i^n + T_{i-1}^n) \quad (\text{B.2})$$

At the interface between layers, the continuity of heat flux has been imposed through local energy balances. In order to impose it, Neumann boundary conditions have been used at nodes. In this way, a heat removed by conduction from the last node is used as a source term for the adjacent one. In this condition, the equation of conduction is in conservative form.

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \quad (\text{B.3})$$

This equation has been implemented since thermal and physical properties of the materials change at the interface. The cell is intended to be half composed of RSZ and half of Inconel 718. The final equation B.6 has been defined substituting the Fourier's law (B.4, B.5) in equation B.3.

$$q_{RSZ} \sim k_{rsz} \frac{T_{i-1} - T_i}{\Delta x_{RSZ}}, \quad (\text{B.4})$$

$$q_{alloy} \sim k_{alloy} \frac{T_{i-1} - T_i}{\Delta x_{alloy}}, \quad (\text{B.5})$$

$$T_i^{n+1} = T_i^n + \frac{\Delta t}{C} \left(k_{RSZ} \frac{T_{i-1} - T_i}{\Delta x_{RSZ}} + k_{alloy} \frac{T_{i+1} - T_i}{\Delta x_{alloy}} \right) \quad (\text{B.6})$$

The thermal capacity C is defined by equation B.7.

$$C = \frac{1}{2}(\rho_{RSZ}c_{p,RSZ}\Delta x_{RSZ} + \rho_{alloy}c_{p,alloy}\Delta x_{alloy}) \quad (\text{B.7})$$

In explicit methods, stability is crucial. Indeed, it is necessary to apply a condition to avoid the timestep becoming too high with respect to the space interval, increasing round-off errors. In this specific situation, the von Neumann criterion has been applied. von Neumann criterion is typically used for constant coefficients linear finite differences schemes. Nevertheless, it can be extended to non-linear systems assuming local linearity and frozen coefficients [41]. The method consists in assuming the space discretization and compute the allowed time interval in order to guarantee stability by applying equation B.8.

$$\Delta t \leq \frac{\Delta x^2}{2\alpha} \quad (\text{B.8})$$

This condition imposes that the Fourier's number is always lower than 0.5. In order to be more conservative, a safety factor of 0.8 has been used.

When this condition is satisfied, the method is considered to be conditionally stable. To obtain the thickness of the RSZ layer, the bisection method has been implemented. The objective of the method is to minimize a function in order to obtain an unknown that in this case is the thickness.

The function to minimize is defined in equation B.9.

$$f(s) = T_{int}(s) - T_{target} \quad (\text{B.9})$$

It considers that, at the interface, the temperature has to be lower than the temperature of decomposition of the case material. This function is compared to a tolerance of $1e-3$. Then, the script defines a value of thickness of the TBC layer. When the previous condition is satisfied, the thickness is assumed to be the correct one.

Figure B.1 shows the convergence of the bisection method.

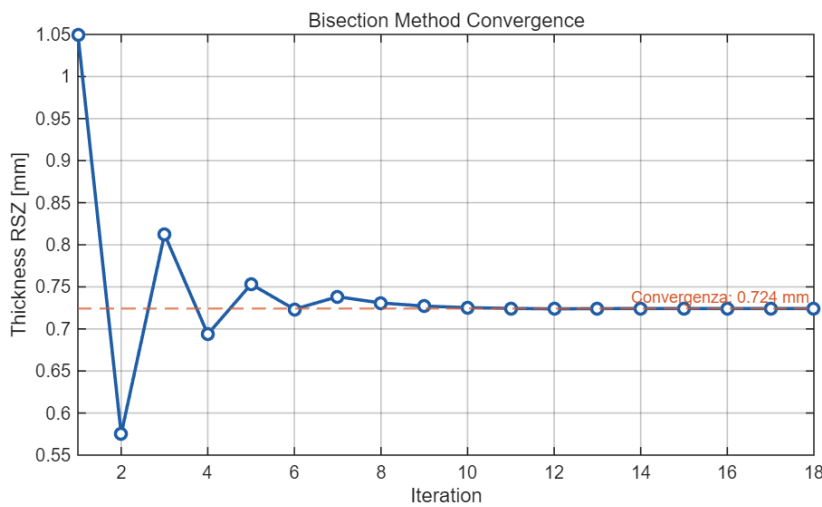


Figure B.1: Bisection method convergence

At the eighteenth iteration, the method stabilizes to the chosen thickness for the TBC.

Appendix C

Monte Carlo uncertainty

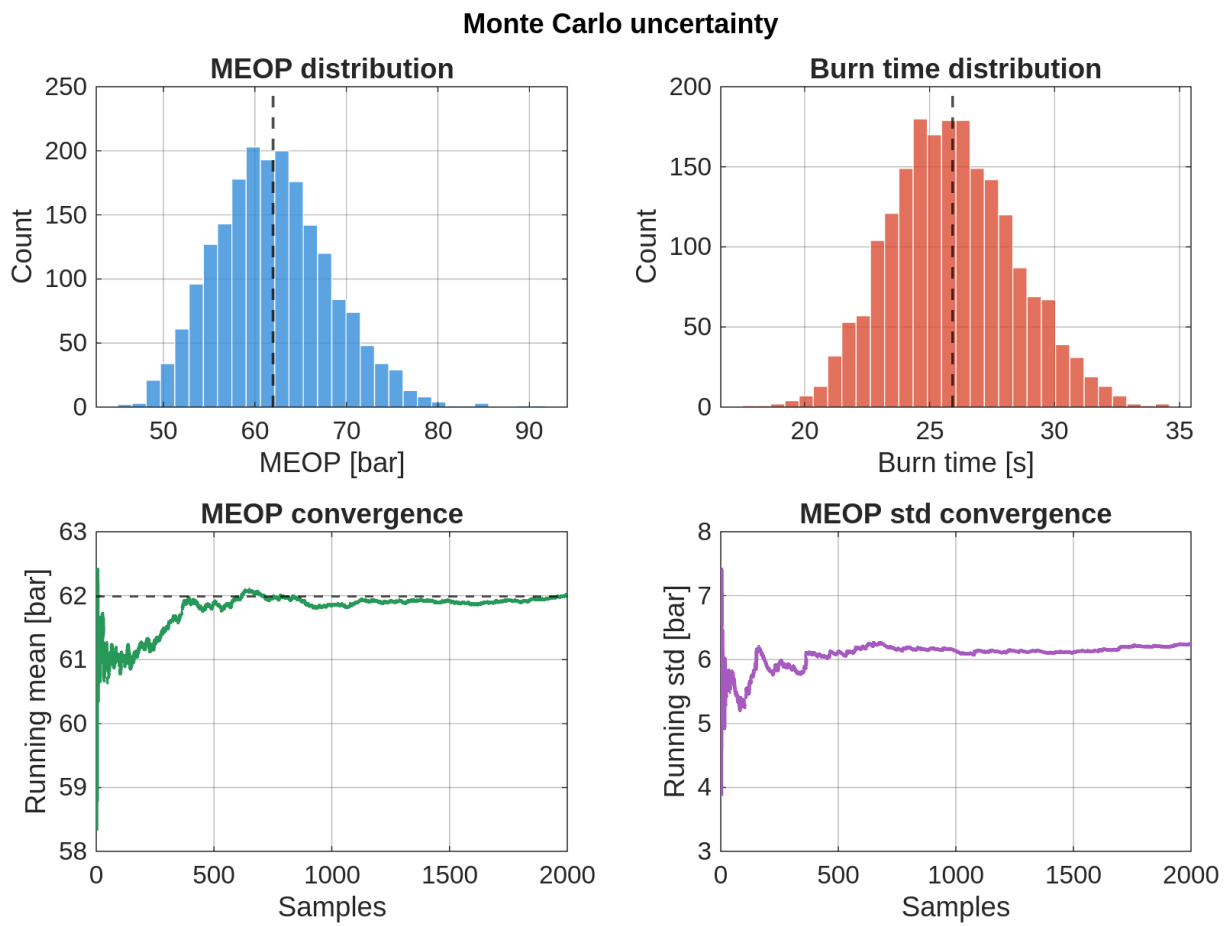


Figure C.1: Monte Carlo Distribution and Convergence Over Study Parameters

Appendix D

Coolant velocity required

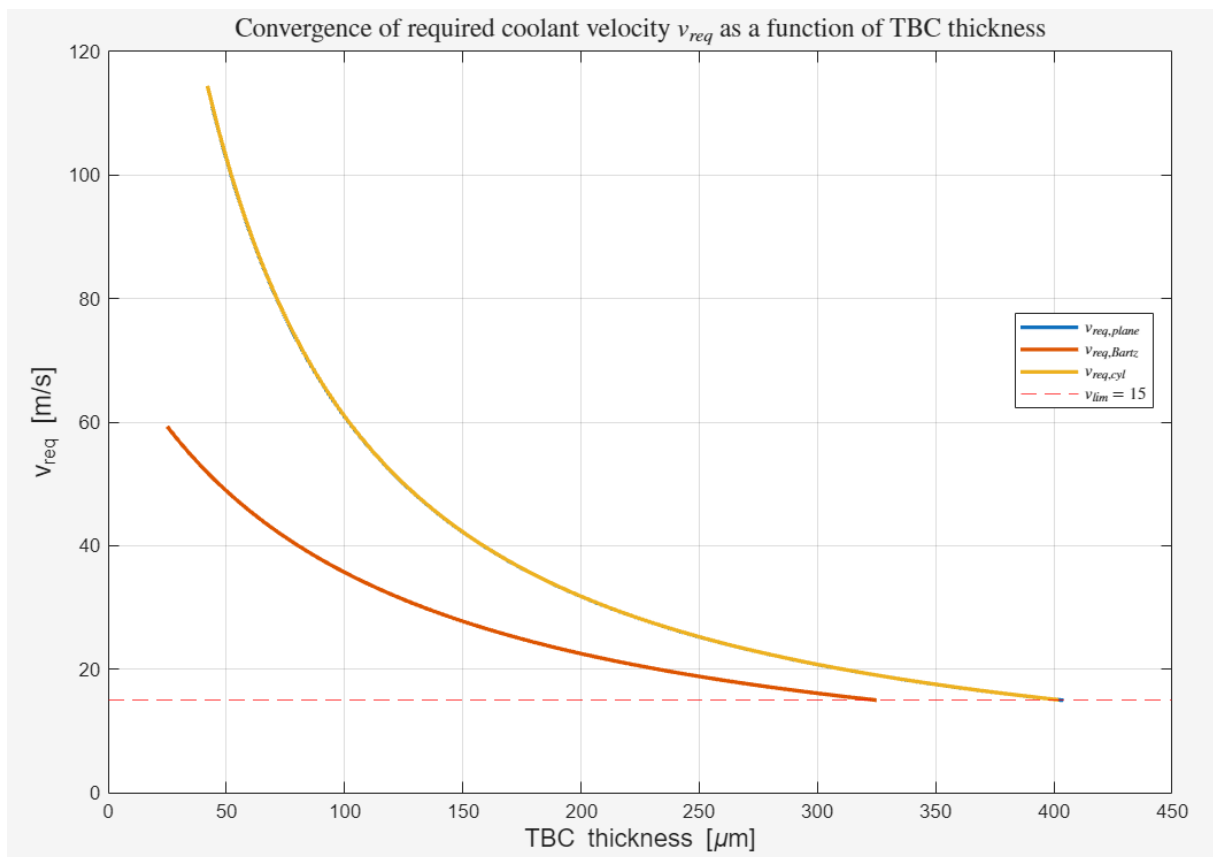


Figure D.1: Coolant velocity required as a function of the TBCs thickness

Appendix E

Profile of the metal/gas interface temperature

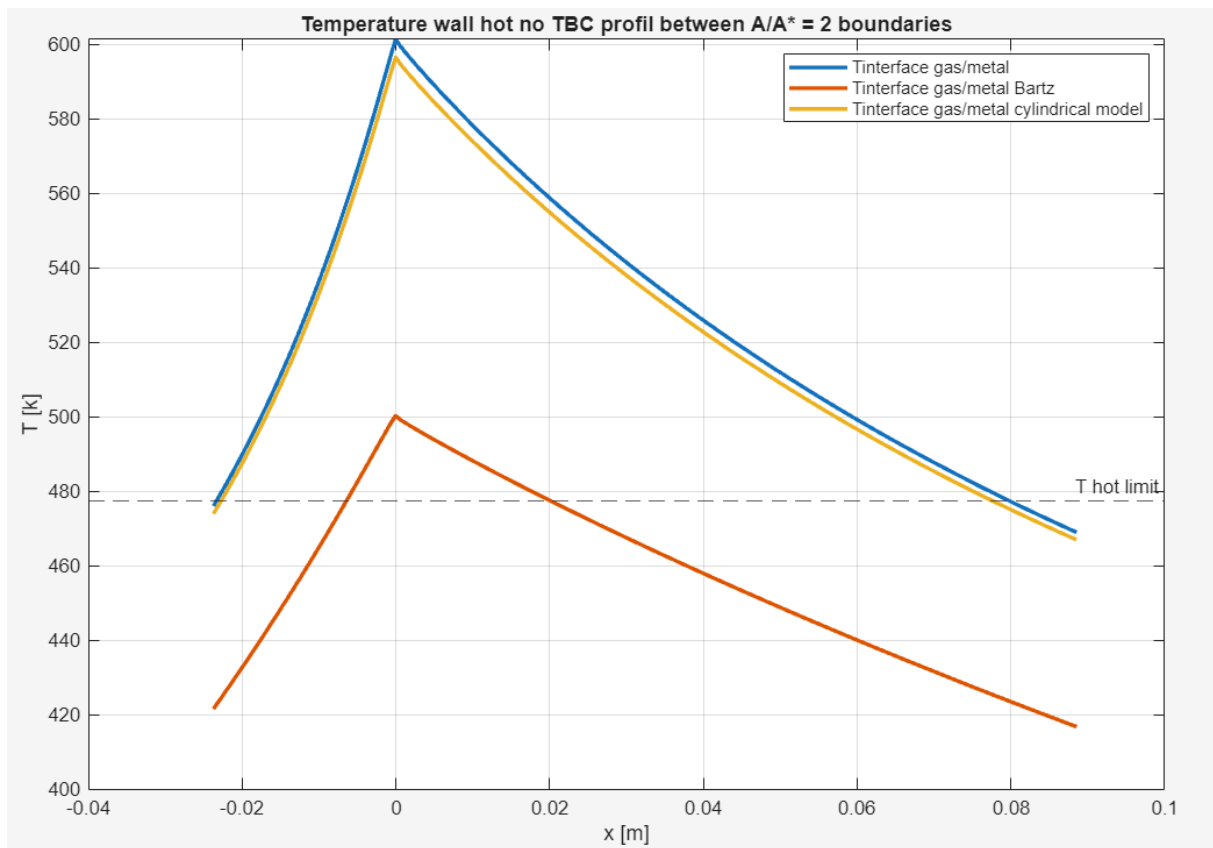


Figure E.1: Profile of the metal/gas interface temperature over the domain